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Type Theory and Categorical Models
A Unified Approach

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Title	Type Theory and Categorical Models: A Unified Approach

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A los «Informales», que han hecho de este máster una experiencia inolvidable y llena de momentazos.
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—Gracias.

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Abstract

This work investigates the interplay between λ-calculus, type theory, and category theory. It is written for readers without a deep mathematical background who nevertheless seek a rigorous but accessible introduction. The exposition emphasizes λ-calculus both as a model of computation and as the foundation for modern type systems, discussing essential properties such as the Church–Rosser theorem and normalization.

Typing disciplines are explored in detail, showing how types act as logical propositions and programs as their proofs, in accordance with the Curry–Howard correspondence. Type theory is then connected to category theory via the structure of cartesian closed categories, showing how categorical semantics provide a unifying perspective.

Throughout, attention is given to why the λ-calculus is a suitable model of computation, how its connection to types and categories enriches our understanding, and how these ideas influence the design of programming languages and related areas, in line with the broader Curry–Howard–Lambek correspondence.

Este trabajo investiga la relación entre el cálculo λ, la teoría de tipos y la teoría de categorías. Está escrito para lectores sin una formación matemática profunda, pero que aun así buscan una introducción rigurosa y accesible. La exposición pone énfasis en el cálculo λ tanto como modelo de computación como fundamento de los sistemas de tipos modernos, discutiendo propiedades esenciales como el teorema de Church–Rosser y la normalización.

Las disciplinas de tipado se exploran en detalle, mostrando cómo los tipos actúan como proposiciones lógicas y los programas como sus pruebas, de acuerdo con la correspondencia de Curry–Howard. Posteriormente, la teoría de tipos se conecta con la teoría de categorías a través de la estructura de las categorías cartesianas cerradas, mostrando cómo la semántica categórica ofrece una perspectiva unificadora.

A lo largo del texto se presta atención a por qué el cálculo λ es un modelo adecuado de computación, cómo su vínculo con los tipos y las categorías enriquece nuestra comprensión, y de qué manera estas ideas influyen en el diseño de los lenguajes de programación y áreas relacionadas, en el marco de la más amplia correspondencia de Curry–Howard–Lambek.

Keywords: Lambda Calculus, Type Theory, Category Theory, Intuitionistic Logic, Curry-Howard-Lambek Correspondence, Closed Cartesian Categories.

Introduction

This work means to provide an introduction to functional programming and categorical reasoning about programs, using the λ-calculus as its foundational formalism for the study of computation. With this in mind, we cover confluence and normalization properties to an acceptable level of detail, as to provide the reader with enough evidence that this system is solid enough.

In its untyped form, the λ-calculus lacks the structure necessary to guarantee properties such as safeness and termination. This lack of structure, naturally leads into typing. We cover monomorphic type systems and the notion of *"programs as proofs"* as introduced by Howard 1980. Typing disciplines enrich the λ-calculus by imposing constraints on expressions, ensuring that only well-formed programs can be constructed, guaranteeing termination and safety. Type systems also give compositional structure to programs, such that given $g : A \to B, f : B \to C$ we can construct $f \circ g : A \to C$. The meaning of this composition is entirely determined by the meanings of f and g and the rule of composition. Compositionality shifts the focus from specific objects to the morphisms between them, and is arguably the core idea behind functional programming.

This principle—where the whole is defined by its parts and their composition—extends beyond functions between sets. Category theory abstracts this idea: it generalizes functions to morphisms and provides a unified language for compositional structure. The work by Lambek 1980 motivates the final chapters of this work, presenting Closed Cartesian Categories, and how they extend the Curry–Howard Isomorphism, giving a categorical interpretation of the work by Curry and Howard.

The following is an informal but approachable summary to this work. This summary is intended as an overview of the landscape connecting type theory and category theory. This is because while we cover the core computational properties of the λ-calculus, the final goal of the document is to make some higher-level connections. To this end, this work emphasizes computation first, and then gradually abstracts toward categorical perspectives as the exposition develops. To do justice to the title, the summary presents the higher-level abstractions that connect the λ-calculus to type theory and category theory. So that the reader understands the discussion going on behind the concrete notions of computation and have an overview of the increasingly abstract perspectives.

Summary

In order to get acquainted with the λ-calculus, let us develop a simple example to familiarize ourselves before we begin with a more formal approach to this discipline. Consider the function $f(x) = x + 1$, the most straightforward way to express this using λ-calculus would be $(\lambda x.x + 1)$, where the lambda denotes that x is being captured and used as a parameter to perform some computation. Evaluating $f(2)$ using this newly created lambda term would look like this: $(\lambda x.x + 1)(2) \rightsquigarrow (2 + 1) \rightsquigarrow 3$. This process of reducing a λ-expression is referred to as β-reduction.

The untyped λ-calculus is extremely expressive but gives no static guarantees about program behaviour — in particular, divergent terms exist, like the Ω combinator 1.3. To regain discipline we annotate terms with types and move to the simply typed λ-calculus. We start from a collection of basic types $A, B, C, \ldots$. From these we form arrow types:

$$\text{if } A \text{ and } B \text{ are types, then } A \to B \text{ is a type}$$

Intuitively, a term of type $A \to B$ is a function which, given an input of type A, returns an output of type B. Because reduction is only performed on well-typed applications, types enforce that compositions are safe: if $M : A \to B$ and $N : A$, then $M\,N$ has type B. For example,

$$\lambda x.\, x + 1 : \mathbb{N} \to \mathbb{N} \qquad (\lambda x.\, x + 1)\, 2 \rightsquigarrow 3$$

so the domain is $\mathbb{N}$ and the codomain $\mathbb{N}$. In short: annotating terms with arrow types restricts how functions can be combined and yields a safer, more predictable calculus. This idea of types can be further extended. The extensions we cover in depth are:

- A product type $A \times B$ in the simply typed λ-calculus means "having both an A and a B at the same time". If $a : A$ and $b : B$, then $\langle a, b\rangle : A \times B$. A product is equipped with projections $\pi_1\langle a, b\rangle = a$, $\pi_2\langle a, b\rangle = b$. In Haskell:

```
data Pair a b = Pair a b -- Product Type: A x B

pi1 :: Pair a b -> a -- Projections
pi1 (Pair x _) = x
pi2 :: Pair a b -> b
pi2 (Pair _ y) = y
```

  Intuitively, this is just like a tuple where π_1 picks the first component and π_2 picks the second.

- In contrast, a sum type $A + B$ in the simply typed λ-calculus means "having either an A or a B, and knowing which one it is". Introduction: if $a : A$ then $\iota_1 a : A + B$; if $b : B$ then $\iota_2 b : A + B$. A case analysis that handles both possibilities:

$$\text{case } s \text{ of } \iota_1(a) \mapsto M \mid \iota_2(b) \mapsto N$$

  Intuitively, this is a tagged union: a value carries a label and a payload. In Haskell:

```
data Sum a b = InL a | InR b -- Sum type: A + B

inl :: a -> Sum a b -- Injections
inl = InL
inr :: b -> Sum a b
inr = InR
```

This extensions allow similar behavior to algebraic types in Haskell, the key difference is that Haskell also allows type definitions, where a type can be defined in terms of itself. The simply typed λ-calculus with sums and products does not have this. For example, lists in Haskell are recursive:

```
data List a = Nil | Cons a (List a)
```

This says a list is either empty or a pair of an element and another list. Such recursive types go beyond what can be expressed with just sums and products in the simply typed λ-calculus and what we will cover.

The simply typed λ-calculus with products and sums has a direct logical correspondence where types correspond to propositions, and terms correspond to proofs. This interpretation is called the Curry–Howard isomorphism. The product type $A \times B$ behaves like logical conjunction $A \wedge B$, and the sum type $A + B$ behaves like logical disjunction $A \vee B$. Establishing that programming with functions and data is formally the same as reasoning in propositional logic.

Up until now we have summarized the role behind the type theoretic grounds that this work covers and how it connects to logic, but this work is intended as a "Unified Approach" to type and category theory. We follow up by informally presenting what categories are and some high level notions of how they relate to types.

Category theory studies objects indirectly, rather than probing the internal nature of an object, one examines how it relates to others, what arrows it admits, and how these compose. In spirit, this is not far from the proverb *"dime de quiénes te rodeas y te diré quién eres"*.

Just as types discipline how terms can be combined in the simply typed λ-calculus, categories discipline how objects can be related through arrows also called morphisms. A category consists of: a collection of objects, a collection of arrows or morphisms between objects and a few important rules about composition. A familiar example is the category **Set**, where objects are sets, arrows are total functions between sets, composition is ordinary function composition.

To begin shifting the readers focus from objects to morphisms, it is helpful to see how familiar notions like injectivity can be re-expressed without referring to elements at all. In the category **Set**, one usually defines a function as injective by saying that distinct elements of the domain are always mapped to distinct elements of the codomain. Category theory instead abstracts this idea: an arrow $f : A \to B$ is called a monomorphism if for any pair of arrows $g_1, g_2 : C \to A$

$$f \circ g_1 = f \circ g_2 \quad \Longrightarrow \quad g_1 = g_2$$

In other words, once composed with f, different arrows into A remain distinguishable. This captures exactly the behavior of injective functions in **Set**, but it does so purely in terms of morphisms and composition. The advantage of this perspective is that it applies uniformly to any category, even in contexts where it makes no sense to speak about individual elements of objects.

In the same way that injectivity can be reformulated in terms of morphisms, constructions like pairs and disjoint unions are captured categorically by the notions of product and coproduct, which embody the behavior of products and sums we covered in typing. Beyond sums and products, categories also support exponentials, which capture the idea of function spaces. Intuitively, an exponential B^A represents "the object of arrows" from A to B. It comes equipped with an evaluation morphism

$$\mathsf{eval} : B^A \times A \to B$$

expressing the idea that a pair of a function and an argument produces a result. This abstractly mirrors how function application works in the λ-calculus.

Putting these ideas together, we arrive at the notion of a Cartesian Closed Category: a category equipped with products, a terminal object, and exponentials. Such categories are themselves deduction systems and provide the natural semantics of the simply typed λ-calculus, where types correspond to objects, terms correspond to morphisms, and function application is captured by the evaluation morphism.

This ongoing interaction between type theory and category theory has inspired the design of entirely new calculi and semantic frameworks. A notable example is the work of Cockett and Gallagher 2019, who developed *Categorical models of the differential λ-calculus* within the setting of differential categories. Their results show how categorical structure can guide the extension of the λ-calculus with new operations such as differentiation, providing both a sound semantics and fresh perspectives on computation. This illustrates how category theory is not merely a language for describing existing systems, but also a tool for discovering and shaping new ones.

Today, the influence of categorical semantics on programming languages is unmistakable. Functional languages such as Haskell, OCaml, and Coq are designed on principles that arise directly from type theory and categorical reasoning, including higher-order functions, algebraic data types, and polymorphism. Monads, for instance, originated as a categorical concept and have become central for structuring effects in modern programming. More recent advances such as dependent types, homotopy type theory, and effect systems show that the dialogue between category theory and type theory continues to inform both the foundations of logic and the design of real programming languages. This ongoing exchange ensures that the state of the art in programming is firmly connected to deep mathematical ideas, with categorical semantics providing both the language and the tools to innovate further.

Contents. As already discussed, this work explores the untyped and typed lambda calculus and the Curry–Howard Isomorphism. We then expand the correspondence by adding categorical semantics to build what is referred to as the Curry–Howard–Lambek Correspondence.

- The first chapter is an introduction to the λ-calculus. It emphasizes confluence as a pillar for computation, it also introduces normalization as a desirable property.
- In the second chapter, typing is introduced, together with its so desired normalization properties. Also, differences with the untyped λ-calculus are discussed.
- The third chapter glances over intuitionistic logic and natural deduction. It introduces the Curry–Howard correspondence.
- The fifth chapter is about category theory and its constructions. We cover functors, natural transformations and adjunctions. We also cover the necessary universal constructions to build Closed Cartesian Categories, namely, terminal objects, products and exponentials.
- The last chapter extends the Curry–Howard correspondence, establishing a relation between λ-calculus and Closed Cartesian Categories.

Target Audience. This text is intended for computer scientists who may be unfamiliar with category theory but are interested in understanding the theoretical foundations of functional programming. It aims to provide readers with enough background on typing so that they can better understand the relation functional programming languages have with category theory, compositionality, type systems and other concepts inside of Category Theory that drive the development of sate-of-the-art programming.

This text assumes basic knowledge of mathematics, familiarity with programming, and a working understanding of recursion. No prior experience with functional programming or category theory is required, though some familiarity would not hurt.

Rationale behind our Approach. This text occupies a middle ground between works like Milewski 2019 and Lane 1971. The former emphasizes intuitiveness, and while it serves as a powerful toolkit for programmers, it gives little insight into the formalisms of categorical reasoning. The latter, presents category theory in full generality, making it inaccessible to non-mathematicians. Our approach introduces the fundamental notions in a somewhat mathematically precise manner first to then develop programmer-friendly examples, applications, and intuitions. This text is not in the same league as the aforementioned, it is a short and concise introduction and should be read as such.

External Resources. The code is meant as a learning aid, nevertheless, it can be given purpose. Similarity to the formal definitions is the driving design principle behind most of the code, as well as ease of use. We use the various programs to simplify some of the examples and avoid including unnecessarily large reduction procedures.

- To clone the LaTeX source used to generate this document:

  ```
  $ git clone https://github.com/laredo02/ttcmua.git
  ```

- To clone the code used to implement the λ-calculus:

  ```
  $ git clone https://github.com/laredo02/lcalc.git
  ```

  There are a few different files in the directory, each of which encapsulates a different approach to typing:

  - Untyped
  - Fully annotated typing
  - Typed with annotations only on abstractions i.e. Typing à la Church
  - Typed with monomorphic type inference i.e. Typing à la Curry

  Files with miscellaneous examples related to categories are also included, in particular those about functors, natural transformations and their composition may be of interest. Implementations are supposed to especially help programmers to digest some of the contents of this work. To use the provided tools, we recommend running them on `ghci`, Haskell's interactive shell.

Disclosure. In the development of this work, I have employed AI tools to assist me with typesetting and at times to find my way with words, as English is not my native tongue. All the content, including explanations, examples, and discussions, is original and developed by me. Whenever external material has been incorporated, it has been cited appropriately. At all times, I have verified the correctness of the information up to my level of competence, ensuring that the final work reflects my own understanding and effort, which is the ultimate goal of the document.

Chapter 1

Untyped λ-Calculus

In the early 20th century, mathematicians such as Bertrand Russell, David Hilbert, and Kurt Gödel were trying to set complete, consistent, and decidable foundations for mathematics. Within this context, Alonzo Church, in his effort to formalize the notion of computability, introduced a minimal symbolic language based upon function abstraction and function application, this system is now called λ-calculus.

In 1936, Church used lambda calculus to address Hilberts Entscheidungsproblem –whether a mechanical method could determine the truth of any first-order logic statement. He proved no such algorithm exists, establishing the undecidability of first-order logic.

One of the nuances formalized by the λ-calculus is the distinction between extensional and intensional equality. The extensional approach to equivalence states that two functions are equivalent if they share input output pairs for every possible input. Its intensional counterpart extends this notion of equality by requiring that the procedures that compute these pairs share complexity i.e. they take the same steps toward yielding a result.[1]

The Lambda Calculus together with Type Theory lay the theoretical groundwork that shaped the design of languages like ML or the more recent Haskell. By itself, the untyped λ-calculus modeled the first versions of Lisp. More advanced type systems than those present in general purpose programming languages enable computers to verify mathematical statements and produce rigorous proofs.

1.1 The Language

Definition 1.1.1. The set of all lambda terms Λ is defined inductively as follows:

- (Variable) If $x \in \mathcal{V}$, then $x \in \Lambda$.
- (Abstraction) If $x \in \mathcal{V}$ and $M \in \Lambda$, then $(\lambda x.M) \in \Lambda$.
- (Application) If $M, N \in \Lambda$, then $(MN) \in \Lambda$.

Where $\mathcal{V} = \{x, y, z, ...\}$ represents a countably infinite set of variable names. These notions are conveniently synthesized into the following abstract syntax:

$$\Lambda := \mathcal{V} \mid \lambda x.\Lambda \mid \Lambda\,\Lambda$$

that directly translates into the following Haskell snippet:

```
data Term = Var String
      | Abs String Term
      | App Term Term
```

The key takeaway of this definition is that abstraction and application when paired with a reduction procedure, capture the meaning of function in a way that is abstract yet useful, and enables computation.

Notation 1. We use uppercase letters, e.g. M, N, L, P, Q to denote elements of Λ and lowercase e.g. x, y, z to denote elements of $\mathcal{V}$.

[1] Let p be a sufficiently large prime, and let $f, g : \mathbb{Z}_p \to \mathbb{Z}_p$ be defined by $f(x) = x^2$ and $g(x) = \log_a(a^{x+2})$, where $a \in \mathbb{Z}_p^\times$ is a fixed primitive root. Although f and g are extensionally equal, i.e., they yield the same output for all inputs in $\mathbb{Z}_p$, they are intensionally distinct. The function f performs a simple squaring operation, while g requires evaluating a discrete logarithm, intractable in general Nadal 2021.

Remark 1. From the definition above we extract that $\lambda x.x{+}1 \notin \Lambda$, given this circumstance, the need to find an accurate representation for numbers within this definition arises. See Definition 1.4.3

Example 1.1.1. Some examples of valid λ terms generated using the syntax from 1.1.1: y, $(\lambda x.(xx))$, $(\lambda x.(\lambda y.x))$, $(((\lambda x.(xy))(\lambda y.y))z)$. The Haskell counterpart for the second term is:

```
Abs "x" (Abs "y" (Var "x"))
```

As a convention to avoid notational cluttering, the outermost parenthesis can be omitted e.g. $\lambda x.x$ is read as $(\lambda x.x)$. Also, application is left-associative and binds tighter than abstraction, so MNP is parsed as $(MN)P$ and $\lambda x.MN$ means $\lambda x.(MN)$, not $(\lambda x.M)N$. Likewise, abstraction is right-associative e.g. $\lambda x.\lambda y.M$ is $\lambda x.(\lambda y.M)$ and binds more weakly than application. Haskell Curry introduced what we now refer to as currying, its main purpose is to avoid unnecessary lambdas. Below is a summary of common notational conventions used throughout the text:

$$\lambda x.x \iff (\lambda x.x) \qquad \lambda x.MN \iff \lambda x.(MN) \qquad MNP \iff (MN)P \qquad \lambda x.\lambda y.f(x,y) \iff \lambda xy.f(x,y)$$

All these are just to keep notation straight-forward, since the formal definition does not leave precedence under-specified thanks to parenthesis. Those interested in the implementation need not worry, as Haskell enforces an explicit order on constructors.

1.2 Reduction of terms

Now that the λ-calculus language has been introduced, it is time we move on to the mechanics of computation, manifested through β-equivalence. α and η equivalence will also be introduced. These two are more difficult to grasp. Informally, it could be said that α-equivalence is equivalence programs up to renaming of local variables. In this same scenario, η-equivalence would singnify equivalence of programs up to wrapping or unwrapping them with an identity function.

Subterms

From the ground up, we define what a subterm is, since the definition of the language is self-referential and terms might contain other terms.

Definition 1.2.1. Given $T \in \Lambda$, $\mathbf{Sub} : \Lambda \to \mathscr{P}(\mathscr{V})$ maps a term to the set of it's subterms.

$$\begin{aligned}
&\mathbf{Sub}(x) = \{x\} \\
&\mathbf{Sub}(MN) = \mathbf{Sub}(M) \cup \mathbf{Sub}(N) \cup \{MN\} \\
&\mathbf{Sub}(\lambda x.M) = \mathbf{Sub}(M) \cup \{\lambda x.M\}
\end{aligned}$$

Where $x \in \mathscr{V}$, $M, N \in \Lambda$ and $\mathscr{P}$ denotes the power set. In Haskell:

```
subTerms :: Term -> Set.Set Term
subTerms (Var x) = Set.singleton (Var x)
subTerms (App m n) =
    Set.insert (App m n) (Set.union (subTerms m) (subTerms n))
subTerms (Abs x m) =
    Set.insert (Abs x m) (subTerms m)
```

Remark 2. Let $\preceq$ be the subterm relation on λ-terms, defined by $M \preceq N \iff M \in \mathbf{Sub}(N)$. $\preceq$ is a partial order relation. This statement is follows intuitively from the definition, it can be proven using induction on the derivation tree.

Definition 1.2.2. A term $S \in \mathbf{Sub}(M)$, $M \in \Lambda$ is said to be proper if $S \neq M$. $\mathbf{Sub}(M) \setminus \{M\}$ would be the set of proper subterms.

Definition 1.2.3. Let $M \equiv \lambda x.N$ be a λ-term:

- The variable x is the binding variable of the abstraction $\lambda x.N$.
- An occurrence of a variable x in M is bound if $x \in \mathbf{Sub}(\lambda x.N)$.
- Free variables are those that are not bound.

Example 1.2.1. Take $M \equiv \lambda x.x\,y$, in this example, the x in $\lambda x.$ is binding, y is free, and x is bound. This means that y is used literally, and that x, as denoted by $\lambda x.$ is abstracted and thus subject to being replaced within the context of said abstraction. A subexample might make this explicit: $(\lambda x.\,x\,y)\,M \rightsquigarrow_\beta M\,y$. The y remained while the x was used as a placeholder for M.

Note 1. This taxonomy is key to understanding concepts such as α-equivalence intuitively.

Free Variables

The set of Free Variables **FV**, those that will not get replaced during the process of reduction, is computed using:

Definition 1.2.4. Given $T \in \Lambda$, $\mathbf{FV} : \Lambda \to \mathscr{P}(\mathscr{V})$ outputs the set of free variables for T:

$$\begin{aligned}
\mathbf{FV}(x) &= \{x\} \\
\mathbf{FV}(MN) &= \mathbf{FV}(M) \cup \mathbf{FV}(N) \\
\mathbf{FV}(\lambda x.M) &= \mathbf{FV}(M) \setminus \{x\}
\end{aligned}$$

Where $x \in \mathscr{V}$ and $M, N \in \Lambda$. In Haskell:

```
freeVars :: Term -> Set.Set String
freeVars (Var x) = Set.singleton x
freeVars (App m n) = Set.union (freeVars m) (freeVars n)
freeVars (Abs x m) = Set.delete x (freeVars m)
```

Example 1.2.2. Computing the free variables of $\lambda x.\lambda y.yxz$. The term well-formed, so we proceed:

$$\begin{aligned}
\mathbf{FV}(\lambda x.\lambda y.yxz) &= \mathbf{FV}(\lambda y.yxz) \setminus \{x\} \\
&= \mathbf{FV}(yxz) \setminus \{x, y\} \\
&= \{x, y, z\} \setminus \{x, y\} \\
&= \{z\}
\end{aligned}$$

Of course, z is the only free variable in the expression, since it is the only variable that is not captured by some λ-abstraction.

Definition 1.2.5. A term $M \in \Lambda$ is closed if and only if $\mathbf{FV}(M) = \emptyset$, closed lambda terms are often referred to as combinators. Λ^0 is the set of all closed lambda terms.

Note 2. They are called combinators since having no free variables means that all variables are bound, and thus, a closed term just combines bound variables.

Variable Substitution

We approach the definition for α-equivalence to this end we have to introduce what it means to substitute a variable.

Definition 1.2.6. Capture-Avoiding Substitution Rules Let $M, N \in \Lambda$ and $x, y, z \in \mathscr{V}$. The substitution $M[x := N]$ is defined inductively as:

$$\begin{aligned}
x[x := N] &= N \\
y[x := N] &= y && y \neq x \\
(M_1\, M_2)[x := N] &= (M_1[x := N])\,(M_2[x := N]) \\
(\lambda x.M)[x := N] &= \lambda x.M \\
(\lambda y.M)[x := N] &= \lambda y.(M[x := N]) && y \neq x, y \notin \mathbf{FV}(N) \\
(\lambda y.M)[x := N] &= \lambda z.(M[y := z][x := N]) && y \neq x, y \in \mathbf{FV}(N)
\end{aligned}$$

where $z \notin \mathbf{FV}(M) \cup \mathbf{FV}(N)$. Implementing the latter in Haskell requires a bit more code than previous definitions, along with making some practical tradeoffs:

```
subst :: String -> Term -> Term -> Term
subst x s v@(Var y)
    | x == y = s
    | otherwise = v
subst x s l@(Abs y body)
    | x == y = l
    | Set.notMember y (freeVars s) = Abs y (subst x s body)
    | otherwise =
        let used = Set.unions [allVars body, allVars s, Set.singleton x]
            z = freshVar used
            body' = rename y z body
        in Abs z (subst x s body')
subst x s (App a b) = App (subst x s a) (subst x s b)
```

Note 3. `freshVar` generates a new name for binding variables such that $z \notin \mathbf{FV}(M) \cup \mathbf{FV}(N)$. More details are available in the source code.

Hopefully the meaning of the first three rules is clear, they ancapsulate renaming in the context of variables and applications. The subsequent rules are about abstractions, rule four states that renaming a binding variable has no effect, renaming only modifies a term whenever the renamed variable is different from the binding variable. Rule five is the general case, that is, when the substitution just moves in together with the abstracted term. Then there is the last case, when the binding variable is a bound occurrence in the term we are substituting, which poses a problem, to see why, let us work through an example:

Example 1.2.3. If we mistakenly substitute x by $(\lambda w.\, w\, y)$ in $(\lambda y.\, y\, x)$ using rule five, we get:

$$(\lambda y.\, y x)[x := (\lambda w.\, w\, y)] = (\lambda y.\, y\, (\lambda w.\, w\, y))$$

This is problematic since the free variable y became bound after the rename. Free and bound variables should maintain their freedom or boundedness after a rename. The correct way to substitute avoiding the name clash:

$$(\lambda y.\, y x)[x := (\lambda w.\, w\, y)] = (\lambda z.\, z\, (\lambda w.\, w\, y))$$

Binding variable serve as placeholders, therefore, renaming should not affect the integrity of an expression.

Note 4. Hence the use of Capture Avoiding alongside Substitution.

One notable result regarding substitution is the substitution lemma:

Lemma 1.2.1. Let M, N, P be lambda terms and x, y be variables. Then:

$$M[x := N][y := P] = \begin{cases} M[x := N[y := P]], & \text{if } x = y, \\ M[y := P][x := N[y := P]], & \text{if } x \neq y \text{ and } y \notin \mathrm{FV}(N). \end{cases}$$

Proof Sketch. By structural induction on M, using α-conversion to avoid capture and assuming bound variables x, y, N, P are fresh.

- Base case: Direct from the definition of substitution; check $z = x$, $z = y$, and z distinct from both.
- Inductive step on applications, where substitution distributes: $(M_1 M_2)[x := N] = M_1[x := N]\, M_2[x := N]$, so both statements follow immediately from the induction hypothesis on M_1 and M_2.
- Inductive step on abstractions with z fresh, $(\lambda z.M_0)[x := N] = \lambda z.(M_0[x := N])$, and the induction hypothesis applies to M_0.

□

In the previous example a bound variable is renamed to avoid a naming collision and avoid binding a previously free variable, α-equivalence is the generalization of this technique, it defines equivalence up to renaming of binding variables.

Notation 2. In many of the upcoming definitions inference rules are used, here is a simple introduction for those unfamiliar with the concept:

$$\frac{P_1 \quad P_2 \quad \cdots \quad P_n}{C}\ \textsc{Name}$$

where P_i are judgments assumed to hold, C is the judgment inferred from the premises and NAME is an optional identification label.

α-equivalence

Definition 1.2.7. Two lambda terms $M, N \in \Lambda$ are α-equivalent, written $M =_\alpha N$, if they are structurally identical except for the names of bound variables. Formally:

$$\frac{}{\lambda x.\, L =_\alpha \lambda y.\, L[x := y]}\ \textsc{Alpha} \qquad \frac{L =_\alpha L'}{\lambda x.\, L =_\alpha \lambda x.\, L'}\ \textsc{Abs}$$

$$\frac{L_1 =_\alpha L_1'}{L_1\, L_2 =_\alpha L_1'\, L_2}\ \textsc{AppL} \qquad \frac{L_2 =_\alpha L_2'}{L_1\, L_2 =_\alpha L_1\, L_2'}\ \textsc{AppR}$$

α-equivalence is an equivalence relation:

$$\begin{array}{ll} \text{(Reflexivity)} & M =_\alpha M \\ \text{(Transitivity)} & M =_\alpha N \text{ and } N =_\alpha P \Rightarrow M =_\alpha P \\ \text{(Symetry)} & M =_\alpha N \Rightarrow N =_\alpha M \end{array}$$

Example 1.2.4. A simple equivalence class and two samples of not α-equivalent pairs.

$$\lambda x.\lambda y.\, y =_\alpha \lambda a.\lambda b.\, b =_\alpha \lambda z.\lambda x.\, x$$

$$\lambda x.\lambda y.\, x \neq_\alpha \lambda x.\lambda y.\, y \quad \lambda x.\, x\, y \neq_\alpha \lambda x.\, x\, z$$

Remember that terms are α-equivalent only when bounded names are properly substituted.

Remark 3. α-equivalence comes naturally once substitution is defined, since the names of bound variables are just placeholders that will eventually hold an argument.

Note 5. Variable naming is a nightmare. De Bruijn indexing is a way of methodically assigning numbers to variable names, indicating how many binders away a variables binder is. This system eliminates the need for α equivalence to exist. See Bruijn 1972.

$$\lambda x.\, x \Rightarrow \lambda.0 \quad \lambda x.\lambda y.\, x \Rightarrow \lambda.\lambda.1 \quad \lambda x.\lambda y.\, y \Rightarrow \lambda.\lambda.0$$

We are better at names than we are at numbers, which is why we will prefer using symbols over De Brujn indexing. Programs like automated theorem provers and compilers do use similar naming strategies to simplify their work.

β-equivalence

Defines the means to perform computation in the λ-calculus, applying substitution as a means of reduction for expressions. This is similar to how one simplifies an arithmetic expression, only it is more powerful, since arithmetic expressions lack expressiveness. It fails to reproduce branching for instance.

Definition 1.2.8. Single-step β-reduction $\rightsquigarrow_\beta$ is defined using the following inference rules:

$$\frac{}{(\lambda x.\, M)\, N \rightsquigarrow_\beta M[x := N]}\ \textsc{Beta} \qquad \frac{M \rightsquigarrow_\beta M'}{\lambda x.\, M \rightsquigarrow_\beta \lambda x.\, M'}\ \textsc{Abs}$$

$$\frac{M_1 \rightsquigarrow_\beta M_1'}{M_1\, M_2 \rightsquigarrow_\beta M_1'\, M_2}\ \textsc{AppL} \qquad \frac{M_2 \rightsquigarrow_\beta M_2'}{M_1\, M_2 \rightsquigarrow_\beta M_1\, M_2'}\ \textsc{AppR}$$

where $M, N \in \Lambda$. In Haskell:

```
step :: Term -> Maybe Term
step (App (Abs x m) n) = Just (subst x n m)
step (App a b) = case step a of
    Just a' -> Just (App a' b)
    Nothing -> case step b of
        Just b' -> Just (App a b')
        Nothing -> Nothing
step (Abs x t) = case step t of
    Just t' -> Just (Abs x t')
    Nothing -> Nothing
step (Var _) = Nothing
```

Single β-reductions can be applied succesively, which induces the following definition

Definition 1.2.9. Multi-step β-reduction. We write $M \rightsquigarrow_\beta^* N$ if and only if there exists a sequence of one-step β-reductions starting from M and ending at N.[2]

$$M \equiv M_0 \rightsquigarrow_\beta M_1 \rightsquigarrow_\beta M_2 \rightsquigarrow_\beta \cdots \rightsquigarrow_\beta M_{n-1} \rightsquigarrow_\beta M_n \equiv N$$

That is, there exists an integer $n \geq 0$ and a sequence of terms $M_0, M_1, \ldots, M_n$ such that: $M_0 \equiv M$ and $M_n \equiv N$ for all i with $0 \leq i < n$, we have $M_i \rightsquigarrow_\beta M_{i+1}$.

[2] The convention in the literature is to use $\rightarrow_\beta$ for β-reductions $\twoheadrightarrow_\beta$ for multi-step β-reductions. We prefer to use $\rightsquigarrow_\beta$ and $\rightsquigarrow_\beta^n$ since this allows us to numerate β-reduction steps e.g. $(\lambda f.\lambda x.\, x)\, M\, N \rightsquigarrow_\beta^2$ N. Instead of $\rightsquigarrow_\beta^n$, we use $\rightsquigarrow_\beta^*$ whenever the number of steps is undefined.

The process of succesively β-reducing a term leads raises the question of whether all terms can be reduced indefinitly. We know the answer to this, since for instance, variables in $\mathcal{V}$ cannot be reduced. The subsequent question is then, Do all all terms have some kind of normal form that cannot be reduced further? This is answered in Section 1.5. We now introduce a formal definition of normal form that will later allow us to gain insight into the normalization properties of the λ-calculus.

Definition 1.2.10. A λ-term M is in normal form if there is no term N such that $M \rightsquigarrow_\beta N$. Equivalently, M contains no subterm of the form $(\lambda x.P)\,Q$ that can be reduced by beta-reduction. We say a term M is irreducible under beta-reduction.

Definition 1.2.11. A reducible expression, or redex for short, is any subterm of the form $(\lambda x.P)\,Q$. Such a term can be reduced by a single beta-reduction step, resulting in $P[x := Q]$.

Definition 1.2.12. β-equivalence relation, equivalence of λ-terms through β-reductions.

$$M =_\beta N \iff M \rightsquigarrow_\beta^* N \vee N \rightsquigarrow_\beta^* M$$

Note 6. $=_\beta$ is the symetric closure of $\rightsquigarrow_\beta^*$.

We now explore a key aspect of the λ-calculus, in the jargon referred to as first-class citizenship, or high-order, a property of terms that restrains them from being exclusively classified as either function or argument. The example below sheds some light on the matter:

Example 1.2.5. High Order by example. Let us β-reduce the λ-term $A\,B\,C$ and inspect the consequences:

$$\begin{aligned}
\overbrace{(\lambda f.\lambda y.f\,f\,y)}^{A}\overbrace{(\lambda x.x+1)}^{B}\overbrace{(2)}^{C} &\rightsquigarrow_\beta (\lambda y.(\lambda x.x+1)(\lambda x.x+1)\,y)(2)\\
&\rightsquigarrow_\beta (\lambda x.x+1)(\lambda x.x+1)(2)\\
&\rightsquigarrow_\beta^* (\lambda x.x+1)(3)\\
&\rightsquigarrow_\beta^* (4)
\end{aligned}$$

Taking a closer look, A, applies f twice to x, as a consequence, B is being applied twice to C. And so, very easily, we have implemented the $x+2$ function via a sequential application of $x+1$ to C.

Remark 4. As seen above, we can pass what we understand as conventional functions as arguments since they are treated alike. One can intuitively apreciate the computational expresivenes this brings with it, and how this syntactic-semantic homogeneity sets the λ-calculus apart from the classical set-theoretic approach to functions.

η-equivalence

This last notion of equivalence, refers to a program being equivalent to itself wrapped with an identity function:

Definition 1.2.13. Two lambda terms M and N are said to be η-equivalent, written $M =_\eta N$, if:

$$\frac{x \notin \mathbf{FV}(M)}{(\lambda x.M\,x) =_\eta M}\ \textsc{Eta} \qquad \frac{M =_\eta M'}{\lambda x.M =_\eta \lambda x.M'}\ \textsc{Abs}$$

$$\frac{M_1 =_\eta M_1'}{M_1\,M_2 =_\eta M_1'\,M_2}\ \textsc{AppL} \qquad \frac{M_2 =_\eta M_2'}{M_1\,M_2 =_\eta M_1\,M_2'}\ \textsc{AppR}$$

The abstraction $\lambda x.(M\,x)$ performs the same operation as M, so if x is not used freely within M, the abstraction serves no purpose. This is shown in Eta, the rest of the inference rules propagate this equivalence throughout the whole term.

Note 7. Two terms are observationally equivalent if they cannot be distinguished by any context that can be applied to them. η-equivalence can be understood as a form of weak extensionality[3], since it does not fully encode observational equivalence.

[3] Understood as we presented it in the introduction

$\beta\eta$-reduction

To later prove the Church–Rosser property, we introduce a set of rules that captures reduction taking extensionality into account, we combine both β and η-reduction into a single relation.

Definition 1.2.14. We write $M \rightsquigarrow_{\beta\eta} N$ if M reduces to N by one application of either a β or an η-rule, possibly in a subterm. The rules are:

$$\frac{}{(\lambda x.M)\,N \;\rightsquigarrow_{\beta\eta}\; M[x := N]}\ \textsc{Beta} \qquad \frac{x \notin \mathbf{FV}(M)}{\lambda x.(M\,x) \;\rightsquigarrow_{\beta\eta}\; M}\ \textsc{Eta}$$

$$\frac{M_1 \rightsquigarrow_{\beta\eta} M_1'}{M_1\,M_2 \;\rightsquigarrow_{\beta\eta}\; M_1'\,M_2}\ \textsc{AppL} \qquad \frac{M_2 \rightsquigarrow_{\beta\eta} M_2'}{M_1\,M_2 \;\rightsquigarrow_{\beta\eta}\; M_1\,M_2'}\ \textsc{AppR}$$

$$\frac{M \rightsquigarrow_{\beta\eta} M'}{\lambda x.M \;\rightsquigarrow_{\beta\eta}\; \lambda x.M'}\ \textsc{Abs}$$

Definition 1.2.15. We write $M \rightsquigarrow^*_{\beta\eta} N$ if M reduces to N by zero or more single-step $\beta\eta$-reductions, the same way we did for β-reduction in 1.2.9.

1.3 Normalization and The Church-Rosser Theorem

In this section we attempt to give the reader an overview of the properties behind this calculus. We analyze what makes the untyped λ-calculus a suitable computation model, and provide enough evidence that it is missing some desirable properties that motivate the introduction of typing.

The Church-Rosser in the λ-Calculus

To this end we specify some tools general to relations, specifically, those that help us reason about confluence and termination.

Definition 1.3.1. Let $R \subseteq A \times A$ be a binary relation on a set A. The reflexive-transitive closure of R, denoted by R^*, is the smallest relation on A such that:

(Reflexivity)	$(x, x) \in R^*$
(Transitivity)	$(x, y) \in R^* \wedge (y, z) \in R^* \Rightarrow (x, z) \in R^*$
(Inclusion of R)	$(x, y) \in R \Rightarrow (x, y) \in R^*$

where $x, y, z \in A$.

Remark 5. R^* contains all pairs that relate to each other in such a way that x can be transformed into y by applying R zero or more times. Essentially, reflexive transitive closures are chains of relations that relate terms from the first to the last of the chain if there is one.

Example 1.3.1. $\rightsquigarrow^*_\beta$ defined in 1.2.9 is the reflexive transitive closure of $\rightsquigarrow_\beta$. Equivalently, $\rightsquigarrow^*_{\beta\eta}$ is too the reflexive transitive closure of $\rightsquigarrow_{\beta\eta}$.

Definition 1.3.2. Let $\rightarrow$ be a binary relation on a set A. The symbol $\rightarrow^*$ denotes reflexive-transitive closure of $\rightarrow$. We define the Church–Rosser poperty:

$$x \rightarrow^* y \wedge x \rightarrow^* z \Rightarrow \exists w \in A,\ y \rightarrow^* w \wedge z \rightarrow^* w$$

where $x, y, z, w \in A$. In a diagram:

$$\begin{tikzcd}
 & x \arrow[dl, "*"'] \arrow[dr, "*"] & \\
y \arrow[dr, dashed, "*"'] & & z \arrow[dl, dashed, "*"] \\
 & w &
\end{tikzcd}$$

Definition 1.3.3. Let $\to$ be a binary relation on a set A. Its reflexive–transitive closure is denoted by $\to^*$. We say that $\to$ has the diamond property if

$$x \to y \wedge x \to z \;\Rightarrow\; \exists w \in A,\; y \to w \wedge z \to w,$$

for all $x, y, z, w \in A$.

The following section covers the proof of this property for the untyped λ-calculus. We deemed it necessary to include the proof since it provides evidence that the λ-calculusis a viable system to perform computation with. Church–Rosser ensures that if a normal form exists, reduction paths can always converge into a unique term. See 1.3.7. This notion of confluence is strengthened when we introduce typing 2.4.

Remark 6. If you have no time to spare, Nagele, Oostrom, and Sternagel 2016 is a mechanized proof of the Church-Rosser Theorem using the Z-property written in Nominal Isabelle, which enables you to swiftly glance over the following section.

Proof Sketch of the Church-Rosser for $\beta\eta$-Reduction

In this section, we prove the Church–Rosser Theorem for $\beta\eta$-reduction as done in Selinger 2008, even though proof is due to Tait and Martin-Löf. We begin by defining a new relation $M \rightrightarrows M'$ on terms, called parallel one-step reduction. It is the smallest relation satisfying the following rules:

$$\frac{P \rightrightarrows P' \qquad N \rightrightarrows N'}{PN \rightrightarrows P'N'}\ \textsc{ParApp} \qquad \frac{N \rightrightarrows N'}{\lambda x.N \rightrightarrows \lambda x.N'}\ \textsc{ParAbs} \qquad \frac{Q \rightrightarrows Q' \qquad N \rightrightarrows N'}{(\lambda x.Q)N \rightrightarrows Q'[x := N']}\ \textsc{ParBeta}$$

$$\frac{P \rightrightarrows P' \qquad x \notin \mathbf{FV}(P)}{\lambda x.(P\,x) \rightrightarrows P'}\ \textsc{ParEta}$$

The relation $\rightrightarrows$ is also reflexive, $x \rightrightarrows x$ for every x.

Remark 7. In this section, we will not provide explicit proofs of the lemmas. Most are straightforward, follow from induction on terms, and their statements are sensible enough to be understood without detailed derivations.

Lemma 1.3.1. For all M, M', if $M \rightsquigarrow_{\beta\eta} M'$ then $M \rightrightarrows M'$.

Lemma 1.3.2. For all M, M', if $M \rightrightarrows M'$ then $M \rightsquigarrow^*_{\beta\eta} M'$.

Lemma 1.3.3. $\rightsquigarrow^*_{\beta\eta}$ is the reflexive, transitive closure of $\rightrightarrows$.

Proof. Section 4.4 from Selinger 2008 proves 1.3.1, 1.3.2, 1.3.3 in full detail. □

The next goal now is to prove that $\rightrightarrows$ satisfies the diamond property, since together with lemma 1.3.3, implies the Church–Rosser Property. To this end, we first define the maximal parallel one-step reduct M^* of a term M as follows:

- $x^* = x$, for a variable.
- $(P\,N)^* = P^*\,N^*$, if $P\,N$ is not a β-redex.
- $((\lambda x.Q)\,N)^* = Q^*[N^*/x]$.
- $(\lambda x.N)^* = \lambda x.N^*$, if $\lambda x.N$ is not an η-redex.
- $(\lambda x.(P\,x))^* = P^*$, if $x \notin \mathbf{FV}(P)$.

Note that M^* depends only on M.

Lemma 1.3.4. Whenever $M \rightrightarrows M'$, then $M' \rightrightarrows M^*$.

Proof. Section 4.4 from Selinger 2008. □

Lemma 1.3.5. If $M \rightrightarrows N$ and $M \rightrightarrows P$, then there exists Z such that $N \rightrightarrows Z$ and $P \rightrightarrows Z$.

Proof. Take $Z = M^*$. □

Theorem 1.3.6. $\rightsquigarrow_{\beta\eta}$ is Church–Rosser.

Proof. Since $\rightrightarrows$ satisfies the diamond property (Lemma 1.3.5), its reflexive–transitive closure $\rightrightarrows^*$ also satisfies the diamond property. By Lemma 1.3.3, $\rightrightarrows^*$ coincides with $\rightsquigarrow^*_{\beta\eta}$. Hence $\rightsquigarrow^*_{\beta\eta}$ satisfies the diamond property, which is exactly the Church–Rosser property for $\rightsquigarrow_{\beta\eta}$. □

Corollary 1.3.7. If a λ-term M reduces to two normal forms N_1 and N_2, then they are unique up to alpha-equivalece:

$$M \rightsquigarrow^*_\beta N_1 \text{ and } M \rightsquigarrow^*_\beta N_2 \text{ and } N_1, N_2 \text{ are normal } \Rightarrow N_1 =_\alpha N_2$$

Proof. By the Church-Rosser theorem, there exists a term P such that $N_1 \rightsquigarrow^*_\beta P$ and $N_2 \rightsquigarrow^*_\beta P$. Since N_1 and N_2 are normal forms, they cannot be reduced further. Thus, the only possible P is $N_1 =_\alpha N_2$. □

Normalization

Normalization properties essentially summarize the termination properties of terms in the context of β-reduction. So, while The Church-Rosser property ensures confluence, normalization focuses on termination i.e. whether reduction eventually terminates.

Definition 1.3.4. Consider a lambda term M and all possible β-reduction sequences

$$M \equiv M_0 \rightsquigarrow_\beta M_1 \rightsquigarrow_\beta \cdots \rightsquigarrow_\beta M_n$$

$$M \equiv N_0 \rightsquigarrow_\beta N_1 \rightsquigarrow_\beta \cdots \rightsquigarrow_\beta$$

$$\cdots$$

We classify M as follows:

- M is strongly normalizing if every such reduction sequence terminates in a normal form. Equivalently, there is no infinite reduction sequence: $M \equiv M_0 \rightsquigarrow_\beta M_1 \rightsquigarrow_\beta M_2 \rightsquigarrow_\beta \cdots$
- M is weakly normalizing if there exists at least one finite reduction sequence that terminates in a normal form: $M \equiv M_0 \rightsquigarrow_\beta \cdots \rightsquigarrow_\beta M_n$
- M is non-normalizing if no reduction sequence starting from M ever reaches a normal form. In this case, every reduction path is either infinite or loops without producing a normal form.

Note 8. Applying our newly found knowledge on Church–Rosser, we know that out of those different reduction sequences, there always exists some other reduction sequence such that the sequences cross paths.

Proposition 1.3.8. A term may be either, weakly normalizing, strongly normalizing or non-normalizing.

Proof. Examples are:

- $\lambda x.x$ is strongly normalizing since it is already in normal form.
- $\Omega = (\lambda x.xx)(\lambda x.xx)$ reduces to itself forever, hence non-normalizing.
- $(\lambda x.y)\,\Omega \rightsquigarrow_\beta y$, so it has a normal form y, but reducing Ω loops, so it is weakly normalizing.

□

Remark 8. Checking whether a term is terminating is undecidable in general, since it is an instance of the halting problem.

Looking at the big picture, we find the Church-Rosser Property to be a non-trivial result that helps us verify that the λ-calculus is a deterministic model of computation. Meaning that if we take a program and run it, it will always give the same output for a given input. Of course, when we say a program, we mean a λ-term, and when we say run, we mean β-reduce. Perhaps the following section enables the reader to better grasp why we informally speak of lambda terms as programs.

Also, we established the untyped λ-calculus to have pretty weak normalization properties, meaning within the space of possible terms we find everything from strongly-normalizing to non-normalizing terms.

1.4 Encodings and Constructions

This section presents a number of conventional constructions that represent key ideas common to computer science and other fields. We are specifically referring to natural numbers, branching statements, booleans and lists among others, as well as the operations over them. These constructions can be used to prove relevant statements concerning the expressiveness of the λ-calculus.

Note 9. We use our Haskell program to bring some computer science constructions to life, which means that some of the details in this section are implementation dependent. In addition, we use the following type aliases to simplify notation: `v = Var; l = Abs; a = App`.

Natural Numbers

We briefly recall the axiomatic characterization of the natural numbers due to Peano. The Peano axioms provide a minimal foundation for arithmetic, based on a constant 0, a successor function S, and induction.

The connection with the λ-calculus arises from the fact that the same structure can be represented purely functionally. By encoding 0, successor, we recover the essential properties of natural numbers without referring to any external notion of arithmetic. This construction, known as Church encoding, illustrates the expressive power of the λ-calculus and provides the groundwork for representing data and computation purely in terms of functions.

Definition 1.4.1. Peano Axioms, the set of natural numbers $\mathbb{N}$

- **(Zero)** $0 \in \mathbb{N}$
- **(Successor)** $\forall n \in \mathbb{N}, S(n) \in \mathbb{N}$
- (Initial) $\forall n \in \mathbb{N}, S(n) \neq 0$
- (Injectivity) $\forall n, m \in \mathbb{N}, S(n) = S(m) \Rightarrow n = m$
- (Induction) $P(0) \wedge \forall (P(n) \Rightarrow P(S(n))) \Rightarrow \forall n, P(n)$

The first four axioms can be summarized using a graph:

Example 1.4.1. Say for instance we mean to represent 3 using Peano notation, using the graph, we would come up with $s(s(s(0)))$, since it is 3 steps away from 0.

Definition 1.4.2. To add and multiply natural numbers we use following the rules:

$$n + m = \begin{cases} n & \text{if } m = 0 \\ s(n + m') & \text{if } m = s(m') \end{cases} \qquad a \cdot b = \begin{cases} 0 & \text{if } b = 0 \\ a \cdot b' + a & \text{if } b = s(b') \end{cases}$$

Definition 1.4.3. Representation of the set of all naturals in the λ-calculus, usually referred to as Church numerals:

$$\overline{0} \equiv \lambda f.\lambda x.x \qquad \texttt{SUCC} \equiv \lambda n.\lambda f.\lambda x.f(nfx)$$

Using our Haskell program, $\overline{0}$ is represented like follows:

```
zero  = l "f" (l "x" (v "x"))
```

Example 1.4.2. The church numerals have been defined similarly to the peano axioms, that is, by bringing into existance an initial element and using the succesor function to define the rest of them. Let us apply `SUCC` a few times to illustrate:

$$\overline{0} \equiv \lambda f.\lambda x.x \qquad \overline{1} \equiv \lambda f.\lambda x.fx \qquad \overline{2} \equiv \lambda f.\lambda x.f(fx) \qquad \overline{\texttt{n}} \equiv \lambda f.\lambda x.f^n x$$

where n denotes the number of times we append f.

Notation 3. We use $\overline{\text{number}}$ to denote a number is church encoded. In our Haskell program we use `nat` to convert from Haskell's prelude type `Natural` to type our custom type `Term`.

```
nat :: Natural :: Term
nat n = normalize (nat_ n)
    where
        nat_ 0 = zero
        nat_ n = a succ_ (nat_ (n - 1))
```

Remark 9. Since λ-terms are first-class citizens, and f and x are abstracted away, the reality is that numbers are a quite general construct that has many distinct applications. Recall Example 1.2.5 where $A \equiv_\alpha \overline{2}$.

Basic Arithmetic

Definition 1.4.4. Now that we have a construction for the naturals, we can define some arithmetic operations:

$$\texttt{ADD} \equiv \lambda m.\lambda n.\lambda f.\lambda x.mf(nfx) \qquad \texttt{MUL} \equiv \lambda f.\lambda x.\lambda f.\lambda x.m(nf)x$$

$$\texttt{ISZERO} \equiv \lambda n.n\,(\lambda x.\texttt{FALSE})\,\texttt{TRUE} \qquad \texttt{PRED} \equiv \lambda n.\lambda f.\lambda x.\big(((n\,(\lambda g.\lambda h.h\,(g\,f)))\,(\lambda u.x))\,(\lambda u.u)\big)$$

PRED computes the predecessor of a church encoded natural. $\texttt{PRED}\,\overline{0} = \overline{0}$. We omit further Haskell implementation of constructions from now on since they are just literal translations to our programs representation of λ-terms.

Example 1.4.3. Now we illustrate how addition works by performing $2+3$ with the system we defined.

$$\begin{aligned}
\texttt{ADD}\,2\,3 &\equiv (\lambda m.\lambda n.\lambda f.\lambda x.m\,f\,(n\,f\,x))\,(\lambda f.\lambda x.f\,f\,x)\,(\lambda f.\lambda x.f\,f\,f\,x)\\
&\leadsto_\beta^* \lambda f.\lambda x.\,(\lambda f.\lambda x.f\,f\,x)\,f\,((\lambda f.\lambda x.f\,f\,f\,x)\,f\,x)\\
&\leadsto_\beta^* \lambda f.\lambda x.\,(\lambda x.f\,f\,x)\,((\lambda x.f\,f\,f\,x)\,x)\\
&\leadsto_\beta (\lambda f.\lambda x.\,f\,f\,f\,f\,f\,x)\\
&\equiv 5
\end{aligned}$$

In Haskell:

```
ghci> normalize (a (a mul (nat 2)) (nat 3))
```

This unequivocally yields `(λf.(λx.f(f(f(f(f(fx)))))))`.

Remark 10. Multiplication works similarly to addition. As for ISZERO, the intuition behind it is that if the number is zero, then $\lambda x.\texttt{FALSE}$ is dropped, and the result is TRUE; otherwise, it is applied at least once, which immediately yields FALSE.

Remark 11. The λ-calculus is at least capable of arithmetic.

Booleans & Branching

The next constructions in line are the truth values of boolean logic, "*true*" and "*false*" for short.

Definition 1.4.5. Boolean truth values: $\texttt{TRUE} \equiv \lambda t.\lambda f.t$ and $\texttt{FALSE} \equiv \lambda t.\lambda f.f$

Remark 12. Note how $\texttt{TRUE}\ M\ N \leadsto_\beta M$ and $\texttt{FALSE}\ M\ N \leadsto_\beta N$

Definition 1.4.6. Conditional, in other words *"if statement"*. $\texttt{IF} \equiv \lambda b.\lambda t.\lambda e.b\,t\,e$

Note 10. IF statement is just a combinator that happens to behave similarly to a branching statement whenever we pass a church encoded boolean as a first argument.

Example 1.4.4. On the correct behabiour of IF i.e. check that it implements branching. $M, N \in \Lambda$ and IF, TRUE, FALSE are as usual:

$$\begin{aligned}
&\texttt{IF}\ \texttt{TRUE}\ M\ N\\
&\equiv (\lambda b.\lambda t.\lambda e.\,b\,t\,e)\ \texttt{TRUE}\ M\ N\\
&\leadsto_\beta (\lambda t.\lambda e.\texttt{TRUE}\ t\,e)\ M\ N\\
&\leadsto_\beta (\lambda e.\texttt{TRUE}\ M\,e)\ N\\
&\leadsto_\beta \texttt{TRUE}\ M\,N \leadsto_\beta^* M
\end{aligned}
\qquad
\begin{aligned}
&\texttt{IF}\ \texttt{FALSE}\ M\ N\\
&\leadsto_\beta^* \texttt{FALSE}\ M\,N \leadsto_\beta^* N
\end{aligned}$$

So the IF just combines b, t, e in such a way that whenever b is a church encoded boolean either t or e is dropped depending on the truth value of b. See Remark 12. From this example we extract that $\lambda x.x$ is equivalent to IF it terms of behaviour, and thus, we could use $\texttt{IF} \equiv \lambda x.x$ interchangeably.

Definition 1.4.7. Using the IF we can implement an universal set of logic gates:

$$\texttt{AND} \equiv \texttt{IF}\,a\,(\texttt{IF}\,b\,\texttt{TRUE}\,\texttt{FALSE})\,\texttt{FALSE} \qquad \texttt{OR} \equiv \texttt{IF}\,a\,\texttt{TRUE}\,\texttt{TRUE}\,(\texttt{IF}\,b\,\texttt{TRUE}\,\texttt{FALSE}) \qquad \texttt{NOT} \equiv \texttt{IF}\,b\,\texttt{FALSE}\,\texttt{TRUE}$$

This allows us to simulate a nand gate, known to be universal. Thus, the lambda calculus is at least equivalent to propositional logic.

Definition 1.4.8. Define the translation $[\![\cdot]\!]$ that translates a formulas from propositional logic to their Church encoded representation:

$$\begin{aligned}
[\![p_i]\!] &= P_i \quad \text{(a } \lambda\text{-variable)} \\
[\![\neg\varphi]\!] &= \texttt{NOT}\,[\![\varphi]\!] \\
[\![\varphi \wedge \psi]\!] &= \texttt{AND}\,[\![\varphi]\!]\,[\![\psi]\!] \\
[\![\varphi \vee \psi]\!] &= \texttt{OR}\,[\![\varphi]\!]\,[\![\psi]\!]
\end{aligned}$$

Proposition 1.4.1. For every propositional formula φ, in particular every conjunctive normal form, there is a λ-term $[\![\varphi]\!]$ built from the variables and the terms $\texttt{AND}, \texttt{OR}, \texttt{NOT}$ such that for each Boolean assignment σ, substituting each propositional variable by $\texttt{TRUE}$ or $\texttt{FALSE}$ yields:

$$\varphi(\sigma) = \mathsf{true} \iff [\![\varphi]\!][\sigma] \rightsquigarrow_\beta^* \texttt{TRUE} \qquad \varphi(\sigma) = \mathsf{false} \iff [\![\varphi]\!][\sigma] \rightsquigarrow_\beta^* \texttt{FALSE}$$

Thus the λ-calculus with these terms can simulate any conjunctive normal form and hence propositional logic.

Proof. Any conjunctive normal form is a formula built from $\neg, \vee, \wedge$, so we can use the representation 1.4.8 that translates from propositional logic in conjunctive normal form into Church encoded terms. We aim to prove the following invariant by structural induction on φ:

$$[\![\varphi]\!][\sigma] \rightsquigarrow_\beta^* \texttt{TRUE} \iff \varphi(\sigma) = \mathsf{true}$$

and the analogous statement for $\texttt{FALSE}$. Where σ is a Boolean assignment and $[\![\varphi]\!][\sigma]$ stands for the substitution that replaces each P_i by $\sigma(p_i) \in \{\texttt{TRUE}, \texttt{FALSE}\}$.

- Base case: If $\varphi = p_i$ then $[\![p_i]\!][\sigma] = \sigma(p_i)$, so the claim holds by definition of σ.
- Inductive step: Assume the claim for φ and ψ for the inductive hypothesis.
  - For negations we have $[\![\neg\varphi]\!][\sigma] = \texttt{NOT}\,[\![\varphi]\!][\sigma]$. By the inductive hypothesis $[\![\varphi]\!][\sigma] \rightsquigarrow_\beta^* \texttt{TRUE}$ if and only if $\varphi(\sigma) = \mathsf{true}$. Since $\texttt{NOT}$ inverts Church Booleans i.e. $\texttt{NOT}\,\texttt{TRUE} \rightsquigarrow_\beta^* \texttt{FALSE}$ and $\texttt{NOT}\,\texttt{FALSE} \rightsquigarrow_\beta^* \texttt{TRUE}$, it follows that $[\![\neg\varphi]\!][\sigma] \rightsquigarrow_\beta^* \texttt{TRUE}$ if and only if $\varphi(\sigma) = \mathsf{false}$, i.e. $\neg\varphi(\sigma) = \mathsf{true}$.
  - For conjunction, $[\![\varphi \wedge \psi]\!][\sigma] = \texttt{AND}\,[\![\varphi]\!][\sigma]\,[\![\psi]\!][\sigma]$. By the inductive hypothesis the two subterms reduce to the Church booleans corresponding to $\varphi(\sigma)$ and $\psi(\sigma)$. The definition of $\texttt{AND}$ ensures it reduces to $\texttt{TRUE}$ exactly when both inputs are $\texttt{TRUE}$, and to $\texttt{FALSE}$ otherwise. Hence if and only if $\varphi(\sigma) = \psi(\sigma) = \mathsf{true}$, i.e. $(\varphi \wedge \psi)(\sigma) = \mathsf{true}$.
  - We apply similar reasoning on disjunctions using $\texttt{OR}$ shows $[\![\varphi \vee \psi]\!][\sigma] \rightsquigarrow_\beta^* \texttt{TRUE}$ iff $\varphi(\sigma) = \mathsf{true}$ or $\psi(\sigma) = \mathsf{true}$.

Thus the invariant holds for all formulas. In particular it holds for conjunctive normal forms, so the translated λ-term evaluates to $\texttt{TRUE}$ exactly when the conjunctive normal form is true under the given assignment. This completes the proof. □

Corollary 1.4.2. Sice the λ-calculus can simulate any conjunctive normal form it is equivalent to propositional logic.

Pairs

A fundamental construct for building data structures is the notion of pairs. They allow us to bind two values together and query them independently.

Definition 1.4.9. $\texttt{PAIR} \equiv \lambda x.\lambda y.\lambda f.f\,x\,y$.

Remark 13. This definition is convenient since $\texttt{PAIR}\,M\,N \rightsquigarrow_\beta^* \lambda f.f\,M\,N$. Now, we can retrieve the first component via $(\lambda f.f\,M\,N)\,\texttt{TRUE}$ and the second via $(\lambda f.f\,M\,N)\,\texttt{FALSE}$. This motivates the following projection operators.

Definition 1.4.10. A $\texttt{PAIR}$ is nothing but an ordered relation $\langle A, B\rangle \in A \times B$. The projections π_1, π_2 are defined as: $\texttt{FST} \equiv \lambda p.p\,(\lambda x.\lambda y.x)$ and $\texttt{SND} \equiv \lambda p.p\,(\lambda x.\lambda y.y)$.

Proposition 1.4.3. A key property of $\texttt{PAIR}$ is: $\texttt{PAIR}\,A\,B \equiv_\beta \texttt{PAIR}\,(\texttt{FST}\,(\texttt{PAIR}\,A\,B))\,(\texttt{SND}\,(\texttt{PAIR}\,A\,B))$.

Lists

Similarly to natural church encoding, lists are treated as a function where: `NIL` selects the base case, and `CONS` builds a new list by applying a function to the head and recursing on the tail. This way, operations on lists can be defined using functions alone.

Definition 1.4.11. We define the empty list in the same spirit as `0` and `FALSE`: $\texttt{NIL} \equiv \lambda f.\lambda z.z$ It represents the terminating element of the inductive definition of lists. To build non-empty lists we use: $\texttt{CONS} \equiv \lambda h.\lambda t.\lambda f.\lambda z.f\, h\,(t\, f\, z)$ which binds a new element h to a pre-existing list t or to `NIL`.

Definition 1.4.12. Some other relevant list operators are:

$$\texttt{ISNIL} \equiv \lambda l.l\,(\lambda h.\lambda t.\texttt{FALSE})\,\texttt{TRUE}$$

$$\texttt{HEAD} \equiv \lambda l.((l\,(\lambda h.\lambda t.h))\,\texttt{NIL}) \qquad \texttt{TAIL} \equiv \lambda l.\lambda c.\lambda n.l\,(\lambda h.\lambda r.\lambda g.g\; h\;(r\; c))\;(\lambda c.n)\;(\lambda h.\lambda t.t)$$

1.5 Fixed Point Combinators and Recursion

We have seen that not all terms are normalizing. The most straight forward way to test this, is to check whether there exists some term that reduces to itself, effectively creating a loop in the β-reduction chain so to speak. In more formal terms this means there exists some M such that $M \rightsquigarrow_\beta^n M$ with $n > 0$. As already seen, $\Omega \equiv (\lambda x.\,x\,x)(\lambda x.\,x\,x)$ fullfills this requirement. Let us recall what goes on while β-reducing this term:

$$\Omega \equiv (\lambda x.\,x\,x)(\lambda x.\,x\,x) \rightsquigarrow_\beta (\lambda x.\,x\,x)(\lambda x.\,x\,x) \rightsquigarrow_\beta^* (\lambda x.\,x\,x)(\lambda x.\,x\,x) \equiv \Omega$$

This so called Ω-combinator, proves that not all terms have a normal form, and shines a beam of light onto how recursion can be achieved in the λ-calculus.

There is one caveat to this, that classically, recursion is achieved through naming, take for instance this definition for the gdc:

$$\underline{\text{gcd}}(a,b) = \begin{cases} a & b = 0 \\ \underline{\text{gcd}}(b,\, a \bmod b) & b \neq 0 \end{cases}$$

Here, when $b \neq 0$ the name gcd is used to refer to the same definition again. In the λ-calculus such self-referencing definitions are not permited since the language does not allow names.

Fixed Points

To bypass this setback the notion of fixed point is of particular interest.

Definition 1.5.1. Let $f : X \to X$ be a function. An element $x \in X$ is called a fixed point of f if $f(x) = x$.

Example 1.5.1. Some simple functions that illustrate what fixed points are:

Function	*Fixed Point(s)*
$f(x) = c$	$x = c$
$g(x) = x$	$\{x \mid x \in \mathbb{R}\}$
$h(x) = x^2$	$x = 0,\, x = 1$

This harmless idea satisfies an interesting property, namely that $x = f(x) = f(f(x)) = f^n(x)$ effectively creating a chain where every element $f^{n+1}(x) = f(f^n(x))$, attaining self-reference without naming and showcasing the ability of fixed points to induce recursion. Our quest now, is to find a term that simulates this behavior. To do this, let us translate the definition for fixed point into the world of λ-calculus.

Definition 1.5.2. Given a lambda term $L \in \Lambda$, a term $M \in \Lambda$ is called a fixed point of L if $M =_\beta LM$.

No surprises here, the definition is close to a one-to-one mapping of its functional counterpart seen previously. Interestingly enough, it turns out that every lambda term has a fixed point:

Theorem 1.5.1. For every $L \in \Lambda$, there exists $M \in \Lambda$ such that M is a fixed point for M i.e. $M =_\beta LM$. Referred to as the Fixed Point Theorem in the literature.

Proof. Given $L \in \Lambda$, define $M := (\lambda x.\, L\,(x\,x))(\lambda x.\, L\,(x\,x))$, since M is reducible:

$$\begin{aligned} M &\equiv (\lambda x.\, L\,(x\,x))(\lambda x.\, L\,(x\,x)) \\ &\rightsquigarrow_\beta L\,((\lambda x.\, L\,(x\,x))(\lambda x.\, L\,(x\,x))) \\ &\equiv L\,M. \end{aligned}$$

Hence, $M =_\beta LM$, as required. □

Remark 14. See how LM is the application of M to L and that using APPR we can keep β-reducing M the same way we did for $x = f(x) = f(f(x))\ldots$ only this time using lambda terms:

$$M \rightsquigarrow_\beta L\,M \rightsquigarrow_\beta L\,L\,M \rightsquigarrow_\beta^* L^n M.$$

Recursive Combinators

Now that nameless self-reference has been proven to exist within the λ-calculus, we take a step forward and introduce a few of the most important fixed point combinators and explore their different natures.

Definition 1.5.3. Y Combinator. This is the term that given some other term returns its fixed point.

$$Y \triangleq \lambda f.\, (\lambda x.\, f\,(x\,x))\,(\lambda x.\, f\,(x\,x))$$

Remark 15. Even though the Y combinator is of theoretical interest since it allows us to find recursion within the λ-calculus, it is of little practical under strict evaluation since it always runs into non-termination e.g.

$$\begin{aligned} Y\,F &\equiv (\lambda f.\, (\lambda x.\, f\,(x\,x))(\lambda x.\, f\,(x\,x)))F \\ &\rightsquigarrow_\beta (\lambda x.\, F\,(x\,x))(\lambda x.\, F\,(x\,x)) \\ &\rightsquigarrow_\beta F((\lambda x.\, F\,(x\,x))(\lambda x.\, F\,(x\,x))) \\ &\rightsquigarrow_\beta F(F((\lambda x.\, F\,(x\,x))(\lambda x.\, F\,(x\,x)))) \\ &\rightsquigarrow_\beta \cdots \end{aligned}$$

Under strict evaluation, the argument F is not evaluated until Y is done unraveling i.e. never.

To work around this issue, several other combinators exist, the Z and the Turing combinators are two of the better known ones:

Definition 1.5.4. $Z \triangleq \lambda f.\, (\lambda x.\, f\,(\lambda v.\, x\,x\,v))\,(\lambda x.\, f\,(\lambda v.\, x\,x\,v))$

Example 1.5.2. Unfolding the Z Combinator

$$\begin{aligned} Z\,F &\equiv (\lambda f.\, (\lambda x.\, f\,(\lambda v.\, x\,x\,v))\,(\lambda x.\, f\,(\lambda v.\, x\,x\,v)))\,F \\ &\rightsquigarrow_\beta (\lambda x.\, F\,(\lambda v.\, x\,x\,v))\,(\lambda x.\, F\,(\lambda v.\, x\,x\,v)) \\ &\rightsquigarrow_\beta F\,(\lambda v.\, (\lambda x.\, F\,(\lambda v.\, x\,x\,v))\,(\lambda x.\, F\,(\lambda v.\, x\,x\,v))\,v) \\ &\equiv F\,(\lambda v.\, Z\,F\,v) \end{aligned}$$

Remark 16. The extra λv in the final form of $Z\,F$ is essential to delay evaluation in a strict environment. Without it, the recursive call would be applied immediately, and cause non-termination in a call-by-value setting.

Definition 1.5.5. Turing combinator $\Theta \triangleq (\lambda x.\, \lambda y.\, y\,(x\,x\,y))\,(\lambda x.\, \lambda y.\, y\,(x\,x\,y))$

We now use the Turing combinator to aid us implement an untyped version of the factorial function.

λ-Calculus Implementation of Factorial

Classically, factorial is defined as $n! = \prod_{k=1}^{n} k$, that has the following recursive interpretation $n! = n \cdot (n-1)!$ with $0! = 1$. This section develops a worked out example of how to implement this in the λ-calculus.

Example 1.5.3. Implementing factorial using the acquired toolkit:

$$\underline{\texttt{fact}} = \lambda n.\, \texttt{if}\ (\texttt{iszero}\ n)\ (\overline{1})\ (\texttt{mult}\ n\ (\underline{\texttt{fact}}\ (\texttt{pred}\ n))),$$

As already seen, the problem with a recursive definition such as this one is that the name `fact`, appears both on the left and the right-hand side. This means finding `fact` requires solving an equation, which we will do thanks to fixed points. We start by abstracting away `fact` from the term:

$$\texttt{fact} = \overbrace{\left(\lambda f.\lambda n.\,\texttt{if}\ (\texttt{iszero}\ n)\ (1)\ (\texttt{mult}\ n\ (f\ (\texttt{pred}\ n)))\right)}^{F}\ \texttt{fact}.$$

Then the last equation becomes $\texttt{fact} = F\,\texttt{fact}$, which is a fixed point equation. We can solve it up to β-equivalence by letting $\texttt{fact} = \Theta F$. Using our Haskell program we define `fact`:

```
fact = a turing
    (l "f" (l "n"
        (a (a (a if_ (a iszero (v "n"))) (nat 1))
        (a (a mul (v "n"))
            (a (v "f")
            (a pred_ (v "n")))))
    ))
```

We run the following command to compute $\texttt{fact}\,\overline{3}$:

```
ghci> normalize (a fact (nat 3))
```

That of course yields $\lambda f.\lambda x.\,f(f(f(f(f(f\,x))))) \equiv \overline{6}$ as result.

Note 11. The latter implementation exhibits a striking syntactic resemblance to that of the Lisp family of languages.

Conclusions

We have covered some of the most important notions that surround the untyped λ-calculus. It has been shown to satisfy the Church-Rosser property of confluence, it has also been shown that this property is not the strongest of its kind. We have also seen that it is not strongly normalizing. For this reasons, it is in our best interest to add some spice to the λ-calculus by means typing. This is to try and make it comply with some stronger requirements that will make it more practical to perform computation. We covered untyped λ-calculus the next chapters focuses on minimal typing variations.

- **Untyped**: Where the type of expressions is never specified. As we have seen, this gives us maximal expressiveness at the cost of being unsafe. Informally, this means we are at a risk of composing unrelated terms that produce miscalculations. For instance xM.
- **Minimal Typing**: Every expression is annotated with a type. This would be equivalent to the usual set theoretic approach to computation where functions have well defined domain and codomain.
- **Extended Typing**: Similar to minimal typing but extended with constructions like products and sums.

This alternatives have dramatically different properties from one another. We will cover them in some detail throughout the text.

References and Resources. This chapter is influenced by many of the classic resources on the subject, primarily Nederpelt and Geuvers 2014 for the basic aspects of the λ-calculus. The proof to the Church-Rosser Theorem and the factorial example are fine-tuned versions of what can be found in Selinger 2008, though, for what it's worth, the factorial implementation is original. Similar proofs are found in O. L. Project 2025 and Nagele, Oostrom, and Sternagel 2016. Other text used as general resources are Henk P. Barendregt and Barendsen 2000 and Girard 1989. We intentionally skipped some of the topics in the literature like combinatory algebras that are not that relevant to our focus of study.

Chapter 2

Typing and Extensions

We covered the untyped λ-calculus, highlighting its strengths and limitations. This section introduces the notion of typing, which restricts its expressiveness while improving the tractability of studying its properties. The idea of typing arises from the usual definition of a function, understood as a mapping between elements of sets. For example, consider a function $f : A \to B$ defined by $a \mapsto f(a)$, where we recognize A as the domain and B as the codomain. Notice how the mapping is meaningful only within this context–otherwise f would lack sense. Hence, the important of restricting what a function can take as an input as well as what it may yield as an output.

In the context of the λ-calculus, types serve as a means of restriction, similar ho how we use logical predicates to define sets. Just as $f : ? \to ?$ is refined to $f : A \to B$ by specifying requirements on the arguments and ensuring properties of the outputs, we use types to impose requirements on the arguments of an abstraction and to ensure properties of the reduction process of that term.

Since the λ-calculus treats all terms as first-class citizens, attempting to impose restrictions directly on arguments is not effective. Instead, we define restrictions on all terms by pairing each term with additional information called type. This tells us about the nature of a given term, which may be of two kinds: a simple type or arrow type $\to$. Introducing a typing system into the λ-calculus splits Λ into two categories: Typable and Non-typable terms. These sets are disjoint. This simple type system classifies expressions such as $x(\lambda y.y)$ or Y as untypable, while $\lambda x.x$ has type $\alpha \to \alpha$. Typing ensures that terms are meaningful: only normalizing terms receive a type, while non-normalizing or ill-defined terms do not.

Out of the many type systems developed for the λ-calculus, we focus on a small but expressive core. We begin with minimal typing, which corresponds to the simply typed λ-calculus with only function types. This system captures the essence of typed functional computation: terms are either variables, abstractions, or applications, and their types are built solely from base types and arrows.

We then extend this core with product types and sum types, which allow richer data structures. Product types model tuples or pairs, supporting construction and projection, while sum types model alternatives, supporting case analysis. These extensions move the type system closer to what is found in practical programming languages, while preserving the foundational simplicity of the λ-calculus.

2.1 Minimal Typing

The untyped version of the λ-calculus relies on abstraction and application. Minimal typing captures this simply by using base types and arrow types.

Definition 2.1.1. The set of minimal simple types $\mathscr{T}$ is defined inductively as follows:

- (Type variable) If $\alpha \in \mathscr{B}$, then $\alpha \in \mathscr{T}$.
- (Arrow type) If $\sigma, \tau \in \mathscr{T}$, then $(\sigma \to \tau) \in \mathscr{T}$.

where $\mathscr{B}$ is a countably infinite set of base types. In abstract syntax:

$$\mathscr{T} ::= \mathscr{B} \mid \mathscr{T} \to \mathscr{T}$$

Note 12. These minimal types are the most basic types one can apply to the λ-calculus, this is with respect to extended types, that use products and sums to add expressiveness.

Notation 4. In general, greek letters $\alpha, \beta, \gamma, \ldots$ are standard to denote type names.

Notation 5. The convention is that $\to$ is right associative, that is: $\alpha \to \gamma \to \alpha$ is the same as $(\alpha \to (\gamma \to \alpha))$.

Example 2.1.1. Examples of simple types include: $\gamma, \quad (\beta \to \gamma), \quad ((\gamma \to \alpha) \to (\alpha \to (\beta \to \gamma)))$

Definition 2.1.2. A context Γ is a finite mapping from variables to simple types, defined inductively by:

$$\Gamma ::= \cdot \mid \Gamma, x : \sigma$$

where

- $\cdot$ is the empty context
- x is a variable
- $\sigma \in \mathscr{T}$ is a simple type
- $\Gamma, x : \sigma$ extends Γ by mapping x to σ
- $\mathrm{dom}(\Gamma) = \{x \mid x : \sigma \in \Gamma\}$
- $\mathrm{rng}(\Gamma) = \{\sigma \mid x : \sigma \in \Gamma\}$

A context is often represented as a finite sequence: $\Gamma = x_1 : \sigma_1, x_2 : \sigma_2, \ldots, x_n : \sigma_n$ with all x_i distinct.

With this in mind we distinguish two alternative methods for typing. Typing à la Church, where types are annotated alongside λ-terms. And Typing a la Curry, where typing judgements are provided externally through some inference algorithm. Church's style of typing adds syntax on top of the already defined λ-calculus. It can be said that while Church's typing introduces rules to generate well-typed terms, Curry's approach infers types from untyped expressions.

- Church-style: $\lambda x : \sigma . x \vdash \lambda x : \sigma . x : \sigma \to \sigma$.
- Curry-style: $\lambda x . x \vdash \lambda x . x : \sigma \to \sigma$.

Note how Curry's style allows us to assign different valid types to terms. For example, $(\alpha \to \beta) \to (\alpha \to \beta)$ could be a valid type assignment for $\lambda x . x$, do not confuse this with polymorphism.

Remark 17. The distinction between Church-style and Curry-style typing has practical consequences for implementation. In Church-style, every abstraction carries its type annotation explicitly, so type checking is straightforward. In Curry-style, terms are written without annotations, and type information must be reconstructed by solving constraints generated during typing. This requires the introduction of type variables and unification algorithms in order to infer consistent typings. Hence Curry-style systems motivate the development of type inference mechanisms, while Church-style systems rely on explicit annotations.

Typing à la Church

Church-style typing embeds type annotations directly in terms, making typing syntax-directed and explicit. We define the following rules that tell whether a term is well-typed.

Definition 2.1.3. We define the following typing relation:

$$\frac{x \notin \mathrm{dom}(\Gamma)}{\Gamma, x : \tau \vdash x : \tau}\ \textsc{Var} \qquad \frac{x \notin \mathrm{dom}(\Gamma) \quad \Gamma, x : \sigma \vdash M : \tau}{\Gamma \vdash \lambda x . M : \sigma \to \tau}\ \textsc{Abs}$$

$$\frac{\Gamma \vdash M : \sigma \to \tau \qquad \Gamma \vdash N : \sigma}{\Gamma \vdash M N : \tau}\ \textsc{App}$$

Example 2.1.2. In this example, we verify that $(\lambda f . \lambda x . f\, x)(\lambda y . y)$ is a valid term of type $\alpha \to \alpha$ using Church-style typing.

$$\dfrac{\dfrac{\dfrac{\dfrac{}{\Gamma, f : \alpha \to \alpha, x : \alpha \vdash f : \alpha \to \alpha}\ \textsc{Var} \quad \dfrac{}{\Gamma, f : \alpha \to \alpha, x : \alpha \vdash x : \alpha}\ \textsc{Var}}{\Gamma, f : \alpha \to \alpha, x : \alpha \vdash f\, x : \alpha}\ \textsc{App}}{\Gamma, f : \alpha \to \alpha \vdash \lambda x . f\, x : \alpha \to \alpha}\ \textsc{Abs}}{\Gamma \vdash \lambda f . \lambda x . f\, x : (\alpha \to \alpha) \to (\alpha \to \alpha)}\ \textsc{Abs} \quad \dfrac{\dfrac{}{\Gamma, y : \alpha \vdash y : \alpha}\ \textsc{Var}}{\Gamma \vdash \lambda y . y : \alpha \to \alpha}\ \textsc{Abs}}{\vdash (\lambda f . \lambda x . f\, x)(\lambda y . y) : \alpha \to \alpha}\ \textsc{App}$$

Church-style annotations allow type inference to be straightforward because every abstraction carries its parameter type explicitly, this comes at the cost of requiring annotations on all binders, making the language more verbose.

Example 2.1.3. The following, is an implementation of types 2.1.3. Then we implement type annotations on abstractions:

```
data Type
    = Base Natural
    | Arr Type Type
data TypedTerm
    = Var String
    | Abs String Type TypedTerm
    | App TypedTerm TypedTerm
```

Then we add the notion of context as a map from variable names to types.

```
type Context = [(String, Type)]
```

We define `infer`:

```
infer :: Context -> TypedTerm -> Maybe Type
infer ctx (Var x) =
  lookup x ctx
infer ctx (Abs x t body) =
  case infer ((x, t) : ctx) body of
    Just bodyType -> Just (Arr t bodyType)
    Nothing -> Nothing
infer ctx (App t1 t2) =
  case infer ctx t1 of
    Just (Arr argType resType) ->
      case infer ctx t2 of
        Just t2Type | t2Type == argType -> Just resType
        _ -> Nothing
    _ -> Nothing

ghci> infer [] (Abs "x" (Base 0) (Var "x"))
Just 0->0
```

Our Church style inference is smart enough to provide us with statements such as:

```
ghci> infer [("y", (Base 0))] (App (Abs "x" (Base 0) (Var "x")) (Var "y"))
Just 0
```

provided that the context carries sufficient information and the expressions are well formed and properly annotated.

Definition 2.1.4. If $\Gamma \vdash M : \sigma$, then we say that M has type σ in Γ. A term M is typable if there exist Γ and σ such that $\Gamma \vdash M : \sigma$.

Typing à la Curry

This typing flavor is probably the more elegant version of the two alternatives presented. It provides complete separation between terms and types, meaning there is no need to extend the syntax of λ-calculus to include type annotations in abstractions. Importantly, in the monomorphic case, if a term is indeed typable, there exists an algorithmic procedure to infer a type for it without requiring explicit type annotations. This approach can be further improved in more advanced systems, sometimes referred to as "Algorithm W", which systematically finds types for typable terms in the polymorphic setting. This idea leads to the Hindley-Milner type system, which allows type inference for terms with polymorphic types.

We implement Typing à la Curry following Nazareth and Nipkow 1996. Algorithm 1 presents a faithful synthesis of their monomorphic version of Algorithm W. We provide a similar alternative that is fully interoperable with the rest of our implementation; however, we omit the code to maintain clarity and readability.

Here, **Var** x and **Abs** e are self-explanatory. For **App** e_1 e_2, the algorithm infers types for the function and argument, unifies them using `mgu`, and returns the resulting type along with the composed substitutions. `TVar(n)` generates fresh type variables, `mgu` computes the most general unifier, and $\circ$ denotes substitution composition.

Algorithm 1 Monomorphic Algorithm W

```
function W(expr, env, n)
    match expr with
      Var x:
    if x ∈ dom(env) then
        return Ok(id, env(x))
    else
        return Fail
    end if
      Abs e:
    (s, t, m) ← W(e, env[x ↦ TVar(n)], n + 1)
    return Ok(s, TVar(n) → t, m)
      App e1 e2:
    (s1, t1, m1) ← W(e1, env, n)
    (s2, t2, m2) ← W(e2, s1(env), m1)
    u ← mgu(s2(t1), t2 → TVar(m2))
    return Ok(u ∘ s2 ∘ s1, u(TVar(m2)), m2 + 1)
end function
```

Example 2.1.4. Using inference algorithms like the one earlier we can provide terms with suitable types without need for explicit type annotations: Let σ, τ, ρ be arbitrary types. Then:

- $\vdash \lambda x.x : \sigma \to \sigma$
- $\vdash \lambda x.\lambda y.x : \sigma \to \tau \to \sigma$
- $\vdash \lambda x.\lambda y.\lambda z.x\,z\,(y\,z) : (\sigma \to \tau \to \rho) \to (\sigma \to \tau) \to \sigma \to \rho$

from our program:

```
ghci> infer [] (Abs "x" (Var "x")) 0
Just ([],(a0 -> a0),1)
ghci> infer [] (Abs "x" (Abs "y" (Var "x"))) 0
Just ([],(a0 -> (a1 -> a0)),2)
```

$\lambda x.\lambda y.\lambda z.x\,z\,(y\,z)$ is omitted for readability. Clearly `a0` corresponds to α and `a1` to τ. Note how the typing context is always empty at first. For testing purposes, let us try and find a type for the Y combinator:

```
ghci> infer [] ycomb 0
Nothing
```

Note 13. Typing style is largely a practical issue, a matter of taste even, since ultimately both approaches yield valid typings. The more we annotate, the less we have to infer, and the converse is also true. For the sake of simplicity, we therefore adopt typing à la Church, keeping type information explicit and straightforward.

Properties of Minimal Typing

In 1.3 we analyzed the untyped λ-calculus as a computation model. We also provided evidence that it is missing some desirable properties, for instance, not all terms reached a normal form, as seen in 1.3, some computations can run forever. This makes reasoning about programs more difficult and motivates the study of typed variants. In the following, we provide a few of the most relevant properties of this minimally typed variant, that wile restricting expressiveness, add safety and decidability. Proofs are omitted, refer to Nederpelt and Geuvers 2014 and H. P. Barendregt 1992.

Definition 2.1.5. The one-step β-reduction relation $\rightsquigarrow_\beta$ on terms of $\Lambda^{\to}$ is the smallest relation satisfying:

- (Basis) $(\lambda x : \sigma.\, M)\, N \rightsquigarrow_\beta M[x := N]$
- (Compatibility) If $M \rightsquigarrow_\beta M'$ then $M\,N \rightsquigarrow_\beta M'\,N, N\,M \rightsquigarrow_\beta N\,M', \lambda x : \sigma.\, M \rightsquigarrow_\beta \lambda x : \sigma.\, M'$

and similarly for all subterms.

Theorem 2.1.1. The ChurchRosser property holds for minimal typing.

Lemma 2.1.2. If $\Gamma \vdash L : \rho$ and $L \to_\beta L'$, then $\Gamma \vdash L' : \rho$.

Theorem 2.1.3. Every well-types term M in minimally typed λ-calculus is strongly normalising.

Remark 18. Proofs to these statements are common-ground among most of the λ-calculus resources cited, see Nederpelt and Geuvers 2014. A proof to the normalization theorem is given 2.4 for the product extension.

2.2 Extended Typing–Products

As discussed, minimal typing is the simplest form of typing. It follows from this statement that it can be extended. Here we introduce products that can be thought of as records or structs:

Definition 2.2.1. Extended types $\mathscr{T}^\times$ is defined inductively as:

$$\mathscr{T}^\times ::= \mathscr{B} \mid \mathscr{T}^\times \to \mathscr{T}^\times \mid \mathscr{T}^\times \times \mathscr{T}^\times \mid 1$$

where $\mathscr{B}$ is the set of base types.

Definition 2.2.2. The set of raw terms $\Lambda^\times$ is defined by:

$$\Lambda^\times ::= x \mid \lambda x : \mathscr{T}.\ \Lambda^\times \mid \Lambda^\times\ \Lambda^\times \mid \langle \Lambda^\times, \Lambda^\times \rangle \mid \pi_1 \Lambda^\times \mid \pi_2 \Lambda^\times \mid *$$

Here:

- $x \in \mathscr{V}$, a countable set of variables
- $\lambda x : \mathscr{T}.\ \Lambda^\times$ is function abstraction with type annotation
- $\langle \Lambda^\times, \Lambda^\times \rangle$ is the term constructor for product values
- π_1 and π_2 are projections on product terms
- $*$ is the unique term of type 1

Definition 2.2.3. Given $t \in \Lambda^\times$, typing judgments are of the form $\Gamma \vdash t : \mathscr{T}$, where Γ is a context $x_1 : T_1, \ldots, x_n : T_n$. The typing rules for $\Lambda^\times$ are:

$$\frac{\Gamma \vdash m : a \to b \qquad \Gamma \vdash n : a}{\Gamma \vdash m\,n : b}\ \textsc{app} \qquad \frac{\Gamma, x : A \vdash M : B}{\Gamma \vdash \lambda x : A.\ M : A \to B}\ \textsc{Abs}$$

$$\frac{\Gamma \vdash M : A \qquad \Gamma \vdash N : B}{\Gamma \vdash \langle M, N \rangle : A \times B}\ \textsc{Pair} \qquad \frac{\Gamma \vdash M : A \times B}{\Gamma \vdash \pi_1 M : A}\ \textsc{Proj}_1 \qquad \frac{\Gamma \vdash M : A \times B}{\Gamma \vdash \pi_2 M : B}\ \textsc{Proj}_2$$

$$\frac{}{\Gamma \vdash * : 1}\ \textsc{Unit} \qquad \frac{}{\Gamma, x : A \vdash x : A}\ \textsc{Var}$$

Reduction Rules

The following presents the core operational semantics for the extended types with products and unit. We naturally extend β-reduction for products and projections. We the do the same for η reduction. It is important to close this root rules under a context to form a full reduction relation.

β-reduction

The core β-reduction rules for simply typed λ-calculus with products and unit are:

$$(\lambda x : A.\ M)\ N \rightsquigarrow_\beta M[x := N]$$

$$\pi_1 \langle M, N \rangle \rightsquigarrow_\beta M \qquad \pi_2 \langle M, N \rangle \rightsquigarrow_\beta N$$

$$\langle \pi_1 M, \pi_2 M \rangle = M$$

$$\lambda x : U.\ (M\,x) = M \quad (x \notin \mathrm{FV}(M))$$

η-laws

The η-laws capture extensionality principles, identifying terms with their canonical expansions:

$$\lambda x : A.\, M\, x \equiv_\eta M \quad (x \notin FV(M))$$

$$\langle \pi_1 M, \pi_2 M \rangle \equiv_\eta M$$

$$M \equiv_\eta * \quad (M : 1)$$

Closure: Unlike β, η is usually treated as an equational rule rather than a reduction. It is automatically closed under contexts, since definitional equality is taken as a congruence.

$\beta\eta$-conversion

The combined theory, called $\beta\eta$-conversion, is the smallest congruence $\equiv_{\beta\eta}$ satisfying:

$$M \rightsquigarrow_\beta N \;\Rightarrow\; M \equiv_{\beta\eta} N \qquad M \equiv_\eta N \;\Rightarrow\; M \equiv_{\beta\eta} N$$

That is, $\beta\eta$-conversion identifies terms up to both computation and extensionality, closed under all term formers.

2.3 Church Rosser Revisited

One important theorem that fails for $\beta\eta$-reduction in the simply-typed λ-calculus with product types is the ChurchRosser property. The culprit is the η_1-rule. For example, let x be a variable of type $A \times 1$, and consider the term

$$M = \langle \pi_1 x, \pi_2 x \rangle.$$

By the $\langle \pi_1 M, \pi_2 M \rangle \equiv_\eta M$ rule, M reduces to x, whereas by the $1 \equiv_\eta$ rule, M reduces to $\langle \pi_1 x, * \rangle$. Both resulting terms are normal forms, so there is no common reduct. Hence, the ChurchRosser property fails:

$$\langle \pi_1 x, \pi_2 x \rangle \xrightarrow{\eta} \langle \pi_1 x, * \rangle \qquad \langle \pi_1 x, \pi_2 x \rangle \xrightarrow{\eta} x.$$

As stated in Selinger 2008, there are two natural ways to restore the ChurchRosser property:

- If we consider only β-reductions and omit all η-rules, confluence holds. Computationally, this is sufficient: β-reductions drive all actual computation, while η-reductions merely simplify terms. Moreover, η-reductions never block β-redexes and always reduce term size. Hence, any β-normal form can be further reduced to a $\beta\eta$-normal form, though not necessarily uniquely.
- Alternatively, if we remove the type 1 and the constant $*$ from the calculus, the ChurchRosser property holds even for $\beta\eta$-reduction.

2.4 Normalization Revisited

The same way we added types and products to the λ-calculus, we now update our previous definition of normal form to adequately adjust to our new requirements:

Definition 2.4.1. A term is normal if none of its subterms is of any of the following forms:

$$\pi_1 \langle u, v \rangle \quad \pi_2 \langle u, v \rangle \quad (\lambda x : u.v)\, u$$

since otherwise it would be reducible.

Remark 19. Definitions for non-normalizing, weakly normalizing and strongly normalizing terms are kept the same, only now we attend to 2.4.1 as definiton of normal form.

Theorem 2.4.1. Every well-typed term in the simply typed λ-calculus with product types and unit type is strongly normalizing. Formally, if $\Gamma \vdash M : A$, then there is no infinite sequence of reductions starting from M:

$$M \rightsquigarrow_\beta M_1 \rightsquigarrow_\beta M_2 \rightsquigarrow_\beta \dots$$

Proof. An machine checked proof for strong normalization in Agda can be found at Urciuoli, Tasistro, and Szasz 2020. □

Corollary 2.4.2. The simply typed λ-calculusis weakly normalizing.

The proofs are lengthy and well known. They are covered in full detail by Girard 1989 chapters 4 and 6. The strong normalization theorem is difficult to prove. In contrast, the proof of the weak normalization theorem is not as hard and provides measure of complexity based on the number of redexes of a given degree within a term.

Proof Sketch of the Weak Normalization Theorem for Product Extension

To prove the weak normalization theorem, we assign a numerical measure of complexity to terms based on the degrees of their redexes. By carefully analyzing how these degrees behave under substitution and reduction, we construct an inductive argument showing that every term can be reduced, step by step, to a normal form

Definition 2.4.2. The degree $\partial(T)$ of a type T is defined as:

- $\partial(T_i) = 1$ if T_i is atomic
- $\partial(U \times V) = \partial(U \to V) = \max(\partial(U), \partial(V)) + 1$

Definition 2.4.3. The degree $\partial(r)$ of a redex 2.4.1 r is defined by:

- $\partial(\pi_1\langle u, v\rangle) = \partial(\pi_2\langle u, v\rangle) = \partial(U \times V)$, where $U \times V$ is the type of $\langle u, v\rangle$
- $\partial((\lambda x.v)\, u) = \partial(U \to V)$, where $U \to V$ is the type of $(\lambda x.v)$

Definition 2.4.4. The degree $d(t)$ of a term t is the supremum of the degrees of the redexes it contains. For any redex r, we have

$$\partial(r) \leq d(r)$$

since $d(r) = \sup\big(\partial(r), \{\partial(r') \mid r' \subset r\}\big)$

Key Lemmas We cover the necessary lemmas to give the reader an intuition of how the proof develops. We omit the proofs since they are already covered in detail by Girard 1989

Lemma 2.4.3. Degree and Substitution: If x is of type U, then

$$d(t[x := u]) \leq \max(d(t), d(u), \partial(U))$$

The effect substitution has on degree does not blow up and has a limited upper bound. Not only that, we also know the upper bound.

Lemma 2.4.4. Degree and Conversion: If $t \to u$, then $d(u) \leq d(t)$. Reduction only decreases or maintains degree.

Lemma 2.4.5. Conversion of Maximal Degree: Let r be a redex of maximal degree n in t, and suppose all redexes strictly contained in r have degree less than n. If u is obtained by converting r to c, then u has strictly fewer redexes of degree n. Meaning that the a term of maximal degree reduces to a term of smaller degree.

Putting the lemmas together, we use the notion of measure to finalize the proof sketch:

Definition 2.4.5. The Measure of a term t, is defined as

$$\mu(t) = (n, m)$$

where: $n = d(t)$ and $m =$ number of redexes of degree n.

Theorem 2.4.6. By iteratively applying Lemma 2.4.5 to reduce the measure $\mu(t)$ in lexicographic order, every term t eventually reaches a normal form. Hence, weak normalization holds.

2.5 Extended Typing–Sums

Sum types can be thought of as disjoint unions, they are the dual to product types, see 4.5.5. Whereas a product type $A \times B$ contains both an A and a B, a sum type $A + B$ contains either an A or a B, together with a tag indicating which side it came from. This makes sum types a natural way to represent alternatives or branching data. As done in Selinger 2008 we only present the extension in the effort avoid being repetitive.

Definition 2.5.1. Extended types $\mathscr{T}^+$ is defined inductively as:

$$\mathscr{T}^+ ::= \mathscr{B} \mid \mathscr{T}^+ \to \mathscr{T}^+ \mid \mathscr{T}^+ + \mathscr{T}^+ \mid 0$$

where $\mathscr{B}$ is the set of base types.

Definition 2.5.2. The set of raw terms Λ^+ is defined by:

$$\Lambda^+ ::= x \mid \lambda x : \mathscr{T}.\ \Lambda^+ \mid \Lambda^+\ \Lambda^+ \mid \iota_1 \Lambda^+ \mid \iota_2 \Lambda^+ \mid \mathsf{case}\ \Lambda^+\ \mathsf{of}\ \iota_1 x \Rightarrow \Lambda^+ \mid \iota_2 y \Rightarrow \Lambda^+ \mid \square_A \Lambda^+$$

Here:

- $x \in \mathscr{V}$, a countable set of variables
- $\iota_1 M$ injects a term of type A into the left summand of $A + B$
- $\iota_2 M$ injects a term of type B into the right summand of $A + B$
- $\mathsf{case}\ M\ \mathsf{of}\ \iota_1 x \Rightarrow N \mid \iota_2 y \Rightarrow P$ performs case analysis on a sum
- $\square_A M$ eliminates an impossible term of type 0, producing a term of type A

Definition 2.5.3. Given $t \in \Lambda^+$, typing judgments are of the form $\Gamma \vdash t : \mathscr{T}$, where Γ is a context $x_1 : T_1, \ldots, x_n : T_n$. The typing rules for Λ^+ are:

$$\frac{\Gamma \vdash M : A}{\Gamma \vdash \iota_1 M : A + B}\ \textsc{Inl} \qquad \frac{\Gamma \vdash M : B}{\Gamma \vdash \iota_2 M : A + B}\ \textsc{Inr}$$

$$\frac{\Gamma \vdash M : A + B \qquad \Gamma, x : A \vdash N : C \qquad \Gamma, y : B \vdash P : C}{\Gamma \vdash \mathsf{case}\ M\ \mathsf{of}\ \iota_1 x \Rightarrow N \mid \iota_2 y \Rightarrow P : C}\ \textsc{Case} \qquad \frac{\Gamma \vdash M : 0}{\Gamma \vdash \square_A M : A}\ \textsc{Abort}$$

Reduction Rules

The operational semantics for sum types naturally extend β-reduction with case distinction and the empty type.

β-reduction

The core β-reduction rules for simply typed λ-calculus with sums and empty type are:

$$\mathsf{case}\ (\iota_1 M)\ \mathsf{of}\ \iota_1 x \Rightarrow N \mid \iota_2 y \Rightarrow P \rightsquigarrow_\beta N[x := M]$$

$$\mathsf{case}\ (\iota_2 M)\ \mathsf{of}\ \iota_1 x \Rightarrow N \mid \iota_2 y \Rightarrow P \rightsquigarrow_\beta P[y := M]$$

$$\square_A M \rightsquigarrow_\beta \text{(no rule, } M : 0 \text{ cannot reduce)}$$

η-laws

The η-laws for sums capture the universal property of coproducts:

$$\mathsf{case}\ M\ \mathsf{of}\ \iota_1 x \Rightarrow \iota_1 x \mid \iota_2 y \Rightarrow \iota_2 y \equiv_\eta M$$

$$\square_A M \equiv_\eta M \quad (M : 0)$$

As with products, η is treated as an equational principle, closed under all contexts.

$\beta\eta$-conversion

The combined theory, called $\beta\eta$-conversion, is the smallest congruence $\equiv_{\beta\eta}$ satisfying:

$$M \rightsquigarrow_\beta N \;\Rightarrow\; M \equiv_{\beta\eta} N \qquad M \equiv_\eta N \;\Rightarrow\; M \equiv_{\beta\eta} N$$

Thus, $\beta\eta$-conversion identifies terms up to computation and extensionality, closed under sums and the empty type.

Remark 20. The term constructors for sums have the following interpretation:

- ι_1 and ι_2 are the canonical injections, marking whether a value belongs to the left or right summand of $A+B$.
- The case construct corresponds to case analysis on a coproduct, analogous to pattern matching in functional programming.
- The term $\square_A$ expresses the unique morphism $0 \to A$, capturing the fact that from an impossible value one may derive a term of any type.

2.6 Extended Typing–Products and Sums

The systems $\Lambda^\times$ and Λ^+ may be combined into a single calculus $\Lambda^{\times+}$ with corresponding type system $\mathscr{T}^{\times+}$. In this setting the formation rules for types, raw terms, typing rules, and reduction rules are the union of those previously introduced for sums and for products. Thus,

Definition 2.6.1. Extended types with products and sums, denoted $\mathscr{T}^{\times+}$, are defined inductively as:

$$\mathscr{T}^{\times+} ::= \mathscr{B} \mid \mathscr{T}^{\times+} \to \mathscr{T}^{\times+} \mid \mathscr{T}^{\times+} \times \mathscr{T}^{\times+} \mid \mathscr{T}^{\times+} + \mathscr{T}^{\times+} \mid 1 \mid 0$$

where $\mathscr{B}$ is the set of base types.

Definition 2.6.2. The set of raw terms with products and sums, denoted $\Lambda^{\times+}$, is given by

$$\Lambda^{\times+} ::= \Lambda^\times \cup \Lambda^+.$$

Remark 21. By uniting sums and products we obtain the full simply typed λ-calculus with finite types. Every finite type can be expressed from 1, 0, sums, and products. The typing and operational rules are exactly those already given, taken together without modification.

Remark 22. Again, η-reduction breaks Church-Rosser. Fixes for this were given in 2.3.

Kinds of problem in Type Theory

These are the usual problems type theorists face. A programmer should be at least familiar with typability, and even more so with Type Checking, the one responsible for the daily headaches of bug finding in typed languages:

- **Typability**: Determine whether a term is legal. Formally, given a term, find a context and type such that $? \vdash \texttt{term} : ?$ holds. If the term is not well-typed, identify where the derivation fails. Type Assignment is a variant where the context is given, so the task is only to find the type: $\Gamma \vdash \texttt{term} : ?$
- **Type Checking**: Given a context, term, and type, check whether $\Gamma \vdash \texttt{term} : \texttt{type}$ holds. The task is purely to verify that the term has the specified type in the given context.
- **Inhabitation**: Given a context and a type, determine whether there exists a term with that type: $\Gamma \vdash ? : \texttt{type}$

Remark 23. In programming languages and theorem provers, these type problems have concrete meanings. Well-typedness asks whether a term or program is valid under the typing rules, ensuring correctness of proofs or programs. Type Checking verifies that a given term has a specified type, allowing compilers or proof assistants to confirm correctness. Term Finding constructs a term of a given type, corresponding to automatic proof generation in theorem provers or program synthesis in programming languages.

Note 14. In our specific case, the implementations given show an increasing level of type inference. The simplest kind is capable of type checking and relies on annotations for every term, the second implements Church–style typing, and the third implements proper inference on monomorphic types given no annotations. The point of this note is to relate type checking and typability through annotations vs. inference, meaning the more type annotations we add, the more typability reduces to checking.

References & Resources Notions that are ubiquitous to Type Theory have been outsourced from either Nederpelt and Geuvers 2014, Selinger 2008 or Sørensen and Urzyczyn 2006. The subsection on normalization is a synthesis of what can be found in Girard 1989. Other texts that might have influenced the development of this section have been cited accordingly. In regard to implementation details, similar implementations to ours have been examined, namely Nazareth and Nipkow 1996, seen before, and Ahmed 2012. The latter covers formal verification of "Algorithm W" and type inference in the **ML** Language.

Chapter 3

Intuitionistic Logic and The Curry–Howard Isomorphism

Unlike classical logic, which is often extended through modal operators or second-order quantifiers, intuitionistic logic takes a different path, not because it adds power, but because it restricts classical reasoning. Classical logic freely allows non-constructive arguments, proofs that assert the existence of an object without providing a way to explicitly construct it. Intuitionistic logic, in contrast, is designed to formalize constructive reasoning, where existence is meaningful only if a specific example can be provided.

3.1 Intuitionistic Logic

To motivate the introduction of Intuitionistic Logic, we present a theorem and two different styles of proof T. O. L. Project 2025:

Theorem 3.1.1. There are irrational numbers a and b such that a^b is rational.

Proof. Let $a, b = \sqrt{2}$

- If $\sqrt{2}^{\sqrt{2}}$ is rational, we are done.
- Otherwise, $\sqrt{2}^{\sqrt{2}}$ is irrational and we can consider $a = \sqrt{2}^{\sqrt{2}}$ and $b = \sqrt{2}$. In which case:

$$a^b = \left(\sqrt{2}^{\sqrt{2}}\right)^{\sqrt{2}} = \sqrt{2}^{\sqrt{2}\cdot\sqrt{2}} = \sqrt{2}^2 = 2,$$

  which is rational so a and b must exist.

□

This is a perfectly valid proof. However, it is somewhat unsatisfying, since it proves existence without exhibiting a specific pair. A more constructive proof is as follows

Proof. Take $a = \sqrt{3}$ and $b = \log_3 4$. Then

$$a^b = \left(\sqrt{3}\right)^{\log_3 4} = 3^{\frac{1}{2}\log_3 4} = \left(3^{\log_3 4}\right)^{1/2} = 4^{1/2} = 2,$$

which is rational. □

Syntax

The syntax for intuitionistic logic is the same as that of propositional logic. Where $\vee, \wedge, \rightarrow$ are used as binary connectives to compose logical statements and $\neg$ is used to negate such statements.

Definition 3.1.1. The set of formulas of propositional intuitionistic logic is defined as follows:

- $\perp$ is an atomic formula.
- Every propositional variable p_i is a formula.
- If φ and ψ are formulas, then $(\varphi \wedge \psi)$ is a formula.
- If φ and ψ are formulas, then $(\varphi \vee \psi)$ is a formula.
- If φ and ψ are formulas, then $(\varphi \rightarrow \psi)$ is a formula.

In addition to the connectives introduced above, the following symbols define the following symbols to simplify notation:

- $\neg\varphi$ abbreviates $(\varphi \rightarrow \perp)$.
- $\varphi \leftrightarrow \psi$ abbreviates $(\varphi \rightarrow \psi) \wedge (\psi \rightarrow \varphi)$.

The Brouwer–Heyting–Kolmogorov Interpretation

The Brouwer–Heyting–Kolmogorov interpretation provides an informal, constructive understanding of intuitionistic connectives. It is based on the notion of a construction, which can be regarded as a constructive proof. The BHK interpretation assigns meanings to the intuitionistic connectives as follows:

- We assume that we know what constitutes a construction for an atomic proposition.
- A construction of $\varphi_1 \wedge \varphi_2$ is a pair $\langle M_1, M_2 \rangle$, where M_1 is a construction of φ_1 and M_2 is a construction of φ_2.
- A construction of $\varphi_1 \vee \varphi_2$ is a pair $\langle s, M \rangle$, where $s = 1$ and M is a construction of φ_1, or $s = 2$ and M is a construction of φ_2.
- A construction of $\varphi \rightarrow \psi$ is a function that converts any construction of φ into a construction of ψ.
- There is no construction for $\perp$; it represents absurdity.
- Defined as $\varphi \rightarrow \perp$. Thus, a construction of $\neg\varphi$ is a function that converts a construction of φ into a construction of $\perp$.

Remark 24. Generally speaking, $\neg\varphi$ means any hypothetical construction φ is impossible of φ yields absurdity $\perp$.

3.2 Natural Deduction

The syntax that to create formulas is not sufficient, we need to pair it with a system that enables the deduction of other formulas from a set of given ones.

Definition 3.2.1. A deductive system is a pair $\mathscr{D} = (\mathscr{A}, \mathscr{R})$, where:

- $\mathscr{A} \subseteq \mathscr{F}$ is a set of axioms and $\mathscr{F}$ is the set of formulas.
- $\mathscr{R}$ is a set of inference rules used to derive new formulas from the already known ones.

A derivation or proof in $\mathscr{D}$ from a set of premises $\Gamma \subseteq \mathscr{F}$ is a finite sequence of formulas $\varphi_1, \ldots, \varphi_n$ such that each φ_i is either:

- an axiom ($\varphi_i \in \mathscr{A}$).
- a member of Γ, or obtained from previous formulas in the sequence by an inference rule in $\mathscr{R}$.

We write $\Gamma \vdash \varphi$ if there exists a derivation of φ from Γ in the deductive system $\mathscr{D}$.

Natural deduction is a deduction system within intuitionistic logic that uses constructive reasoning to derive conclusions. This structured proofs emphasize the existence of explicit constructions rather than abstract truth values.

The rules of natural deduction for intuitionistic propositional logic can be presented as follows:

Introduction / Elimination Rules

$$\frac{\phi \qquad \psi}{\phi \wedge \psi}\ \textsc{IntroConj} \qquad \frac{\phi \wedge \psi}{\phi}\ \textsc{ElimConj-L} \qquad \frac{\phi \wedge \psi}{\psi}\ \textsc{ElimConj-R}$$

$$\frac{\phi}{\phi \vee \psi}\ \textsc{IntroDisj-L} \qquad \frac{\psi}{\phi \vee \psi}\ \textsc{IntroDisj-R} \qquad \frac{\phi \vee \psi \qquad [\phi]^n \ \cdots\ \chi \qquad [\psi]^m \ \cdots\ \chi}{\chi}\ \textsc{ElimDisj}$$

$$\frac{[\phi]^n \ \cdots\ \psi}{\phi \to \psi}\ \textsc{IntroImpl} \qquad \frac{\phi \qquad \phi \to \psi}{\psi}\ \textsc{ElimImpl}$$

$$\frac{[\phi]^n \ \cdots\ \bot}{\neg\phi}\ \textsc{IntroNeg} \qquad \frac{\phi \qquad \neg\phi}{\bot}\ \textsc{ElimNeg}$$

$$\frac{\bot}{\phi}\ \textsc{ExFalso}$$

3.3 Alternative Presentation of Natural Deduction

In the standard presentation of natural deduction, assumptions are introduced as leaves of a proof tree and later discharged dynamically. This makes it somewhat unclear when and where each assumption is canceled in the finished derivation. Texts like Selinger 2008 use an alternative approach to present derivations in terms of judgments, which are of the form

$$x_1 : A_1, \ldots, x_n : A_n \ \vdash\ B.$$

Such a judgment expresses that the formula B can be derived from the assumptions $A_1, \ldots, A_n$, where each assumption is labeled by a distinct variable. Every line in a derivation explicitly carries the full set of assumptions used to obtain it. This makes derivations more verbose, but assumption management becomes fully transparent.

New style of Introduction / Elimination Rules

Let Γ denote a context of labeled assumptions. The rules of natural deduction can be reformulated as follows:

$$\frac{}{\Gamma, x : A \vdash A}\ \textsc{Assume-X} \qquad \frac{}{\Gamma \vdash \top}\ \textsc{IntroTop}$$

$$\frac{\Gamma \vdash A \wedge B}{\Gamma \vdash A}\ \textsc{ElimConj-L} \qquad \frac{\Gamma \vdash A \wedge B}{\Gamma \vdash B}\ \textsc{ElimConj-R} \qquad \frac{\Gamma \vdash A \qquad \Gamma \vdash B}{\Gamma \vdash A \wedge B}\ \textsc{IntroConj}$$

$$\frac{\Gamma \vdash A}{\Gamma \vdash A \vee B}\ \textsc{IntroDisj-L} \qquad \frac{\Gamma \vdash B}{\Gamma \vdash A \vee B}\ \textsc{IntroDisj-R} \qquad \frac{\Gamma \vdash A \vee B \qquad \Gamma, x : A \vdash \chi \qquad \Gamma, y : B \vdash \chi}{\Gamma \vdash \chi}\ \textsc{ElimDisj}$$

$$\frac{\Gamma, x : A \vdash B}{\Gamma \vdash A \to B}\ \textsc{IntroImpl} \qquad \frac{\Gamma \vdash A \to B \qquad \Gamma \vdash A}{\Gamma \vdash B}\ \textsc{ElimImpl}$$

$$\frac{\Gamma, x : A \vdash \bot}{\Gamma \vdash \neg A}\ \textsc{IntroNeg} \qquad \frac{\Gamma \vdash A \qquad \Gamma \vdash \neg A}{\Gamma \vdash \bot}\ \textsc{ElimNeg}$$

$$\frac{\Gamma \vdash \bot}{\Gamma \vdash \chi}\ \textsc{ExFalso}$$

Example 3.3.1. Comparing the two alternative styles

Traditional

$$\dfrac{\dfrac{[A \wedge B]}{A}\ \textsc{ElimConj-L}}{\vdash (A \wedge B) \to A}\ \textsc{IntroImpl}$$

Explicit Contexts

$$\dfrac{\dfrac{\dfrac{}{x : A \wedge B \vdash A \wedge B}\ \textsc{Assume-X}}{x : A \wedge B \vdash A}\ \textsc{ElimConj-L}}{\vdash (A \wedge B) \to A}\ \textsc{IntroImpl}$$

Note 15. Going over Example 3.3.1, we observe:

- In the traditional style, we assume $A \wedge B$ at the leaf, apply ElimConj-L to obtain A, and then discharge the assumption with IntroImpl.
- In the explicit-context style, each judgment carries its assumptions. The axiom Assume-X gives $x : A \wedge B \vdash A \wedge B$, then ElimConj-L gives $x : A \wedge B \vdash A$, and finally IntroImpl discharges the labeled assumption $x : A \wedge B$.

Note 16. In the traditional style, assumptions appear only at the leaves and are canceled dynamically. In the alternative style, contexts are explicitly tracked in every judgment, making assumption management fully transparent at the cost of greater verbosity.

3.4 The Curry–Howard Isomorphism

The λ-calculus, as introduced in the first section, provides a minimal but expressive framework for modeling computation. Independently, intuitionistic logic offers a formal system for constructive reasoning that requires the existence of witnesses for propositions. The Curry-Howard correspondence reveals a structural equivalence between them, providing a unifying perspective that connects proofs with programs. The Curry–Howard correspondence establishes the following associations:

- **Propositions ↔ Types**: Proposition in intuitionistic logic can be interpreted as a type in a typed λ-calculus.
- **Proofs ↔ Programs**: Constructive proofs of a proposition correspond to a λ-term inhabiting the associated type.
- **Proof normalization ↔ Program evaluation**: The process of simplifying proofs corresponds to the process of evaluating a term through β-reduction.

"Proofs as Programs"

Given a proposition P in intuitionistic logic, constructing a proof of P is equivalent to constructing a term t such that $t : T$, where T is the type corresponding to P. This yields the principle often summarized as *"proofs-as-programs"* or *"propositions-as-types"*. The specifics of this relation are summarized in table 3.1.

Table 3.1: The Curry–Howard Correspondence

Intuitionistic Logic	**Typed λ-Calculus**
Proposition A	Type A
Proof of A	Term of type A
Implication $A \to B$	Function type $A \to B$
Conjunction $A \wedge B$	Product type $A \times B$
Disjunction $A \vee B$	Sum type $A + B$
$\top$	1
$\bot$	0
Proof normalization	β-reduction

Example 3.4.1. Consider the implication $A \to B$ in intuitionistic logic. A proof of $A \to B$ is a construction that, given a proof of A, produces a proof of B. Under the Curry–Howard correspondence, this directly maps to a function type $A \to B$ in λ-calculus, where a λ-term $f : A \to B$ takes an argument of type A and returns a result of type B. Similarly, conjunction $A \wedge B$ corresponds to a product type $A \times B$, and disjunction $A \vee B$ corresponds to a sum type $A + B$. In each case, the rules for constructing and deconstructing proofs mirror the operations on the corresponding λ-terms.

Example 3.4.2. Modus Ponens via Curry–Howard. The logical rule for modus ponens is:

$$\frac{P \to Q \quad P}{Q}\,\text{MP}$$

Walking our way back through table 3.1, the λ–term should be of type $(P \to Q) \to P \to Q$, since it takes a proof of $P \to Q$ and a proof of P and produces a proof of Q. Also, we can inhabitate the type using the fact that applying a term of type $(P \to Q)$ to a term of type P yields a term of type Q by APP in the typed λ-calculus. The rule MP corresponds to the following typed term:

$$\vdash \lambda f.\lambda x.\, f\, x : (P \to Q) \to P \to Q$$

Step-by-step β-reduction:

$$(\lambda f.\lambda x.\, f\, x)\ f\ x \ \rightsquigarrow_\beta\ (\lambda x.\, f\, x)\ x \ \rightsquigarrow_\beta\ f\, x$$

This reduction might not say much by itself, so let us annotate it with the most general type of each of the terms in the reduction:

$$\begin{aligned}
&(\lambda f : P \to Q.\lambda x : P.\, f\, x)\ \ f : P \to Q\ \ x : P && \vdash (P \to Q) \to P \to Q \\
&\rightsquigarrow_\beta\ \ (\lambda x : P.\, f\, x)\ \ x : P && \vdash P \to Q \\
&\rightsquigarrow_\beta\ \ f\, x && \vdash Q
\end{aligned}$$

References & Resources. An accessible entry point to intuitionistic logic and natural deduction is given in the lecture notes by Selinger 2008, it is rigorous yet pedagogical. A more modular and systematically updated resource is the Open Logic Project T. O. L. Project 2025. Girard's *Proofs and Types* Girard 1989 has also influenced this chapter, as well as Sørensen and Urzyczyns *Lectures on the Curry–Howard Isomorphism* Sørensen and Urzyczyn 2006, that presents the foundational connections between intuitionistic logic, proof theory, and type theory.

Chapter 4

Categories at Work

Category Theory abstracts the notion of internal structure and studies mathematical objects indirectly. Morphisms are the central objects of study, compositionality and abstraction drive the categorical thought process. In this context, constructions are studied though their relations to one another, instead of from a definitional stand point. This keeps us away from manipulating objects directly, instead, encourages us to apply principles of universality to pinpoint specific properties by uniqueness of morphisms.

It also offers extreme generality and has great power to unify seemingly unrelated fields. It can serve to create blueprints that describe mathematical environments and their properties, and to give structure to patterns that repeat within different fields of mathematics. In some instances, category theory serves as a medium for transporting statements between domains. A categorical formulation some result can often be translated to other context via functors or equivalences of categories.

One field that benefits from practical applications of categorical reasoning is computer science. In particular, categories and type theory are closely intertwined through the notion of compositionality: in category theory, composition is a fundamental structural principle, whereas in type theory, compositionality ensures that complex terms and proofs are constructed in a manner that preserves their meaning.

4.1 Categories: An Overview

For our purposes, categories arise as a tool to study compositionality, which is why its subject of study are morphisms and not objects. From this point of view, defining what a category is should not result unfamiliar:

Definition 4.1.1. A category $\mathscr{C}$ consists of:

- A collection of objects $\mathrm{Ob}(\mathscr{C})$
- A collection of arrows or morphisims where each arrow f has a domain object $\mathrm{dom}(f)$ and a codomain object $\mathrm{cod}(f)$. These can be defined using $f : A \to B$ to signify $\mathrm{dom}(f) = A$ and $\mathrm{cod}(f) = B$
- A composition operator assigning to each pair of arrows f and g, with $cod(f) = dom(g)$, a composite arrow $g \circ f : \mathrm{dom}(f) \to \mathrm{cod}(g)$ satisfying the following associative law:

  $$gh \circ (g \circ f) = (h \circ g) \circ f$$

  where $f : A \to B, g : B \to C, h : C \to D$ are arrows of a category
- For each object A, an identity arrow $id_A : A \to A$ satisfying the following identity law:

  $$\mathrm{id_B} \circ f = f \quad f \circ \mathrm{id_A} = f$$

  for any $f : A \to B$

Remark 25. Literature on the topic usually differentiates between small and large categories. Small categories are those whose collection of objects and whose collection of arrows are both sets. Large categories allow for collections of objects or arrows that are proper classes rather than sets. We will only be concerned with small categories.

Example 4.1.1. Various mathematical constructs fit the description of a category. For example, the category **Set** consists of sets as objects and functions as morphisms, where composition is ordinary function composition and identities are identity functions. Another example is the category **Grp**, whose objects are groups and whose morphisms are group homomorphisms.

Proposition 4.1.1. **Set** is a category where objects correspond to sets $A, B, C, \ldots$ and arrows to functions. For $f : A \to B$ and $g : B \to C$, composition of functions is $(g \circ f)(a) = g(f(a))$, which is a function $g \circ f : A \to C$. Composition is associative since for $h : C \to D$ we have that: $h \circ (g \circ f)(a) = h(g(f(a))) = (h \circ g)(f(a)) = (h \circ g) \circ f(a)$. Lastly, each set A, the identity morphism is $\mathrm{id}_A(a) = a$, satisfying $f \circ \mathrm{id}_A = f = \mathrm{id}_B \circ f$. Therefore, **Set** is a category. □

Hom-set

Take any category $\mathscr{C}$, note how from any given objects $a, b \in \mathscr{C}$, in the general case, there potentially exist several different morphisms. Hom-sets are nothing but the set of all morphisms that can relate any two given objects in a category.

Definition 4.1.2. In a category $\mathscr{C}$, for any two objects $A, B \in \mathrm{Ob}(\mathscr{C})$, the hom-set from A to B is the set of all morphisms from A to B. This set is denoted by $\mathrm{Hom}(A, B)$.

$$\mathrm{Hom}(A, B) = \{f \mid f\colon A \to B \text{ is a morphism in } \mathscr{C}\}$$

We use sets since we deal only with small categories.

Example 4.1.2. Let $\mathscr{C}$ denote a category, consisting of a set of objects, $\mathrm{Ob}(\mathscr{C})$, and a set of arrows, $\mathrm{Hom}(\mathscr{C})$, satisfying the category axioms.

- The empty category, denoted by 0, is the category with no objects and no morphisms.

$$\mathrm{Ob}(0) = \emptyset, \quad \mathrm{Hom}(0) = \emptyset$$

- The point category, denoted by 1, is the category with a single object and a single morphism, which must be the identity morphism on that object.

$$\mathrm{Ob}(1) = \{*\}, \quad \mathrm{Hom}(1) = \{\mathrm{id}_*\}$$

  where $\mathrm{id}_* : * \to *$ is the identity morphism on $*$.

- Category 2, often representing the poset $A \leq B$, is a category with two objects and three morphisms. Formally,

$$\mathrm{Ob}(2) = \{A, B\}, \quad \mathrm{Hom}(2) = \{\mathrm{id}_A, \mathrm{id}_B, f\}$$

  where $\mathrm{id}_A : A \to A$ and $\mathrm{id}_B : B \to B$ are identity morphisms, and $f : A \to B$ is a unique non-identity morphism. The compositions are defined as $f \circ \mathrm{id}_A = f$ and $\mathrm{id}_B \circ f = f$.

Definition 4.1.3. A directed graph consists of a collection of vertices and a collection of edges, where each edge has a designated source vertex and target vertex.

Example 4.1.3. Using a directed graph we can represent the categories 1 and 2 given in 4.1.2, as well as category 3 and further. It goes without saying that we won't use directed graphs to represent the 0 category.

$$\begin{tikzcd}
* \arrow[loop right, "id_*"] & & & B \arrow[loop right, "id_B"] \arrow[d] \\
id_A \arrow[loop left] & A \arrow[r] & B \arrow[loop right, "id_B"] & id_C \arrow[loop left] & A \arrow[r] \arrow[ur] & C \arrow[loop right, "id_C"]
\end{tikzcd}$$

Proposition 4.1.2. Any partially ordered set $(P, \leq)$ can be viewed as a category **Pos** whose objects are elements of P and there is a unique arrow $p \to q$ if and only if $p \leq q$. Composition is defined by transitivity: if $p \leq q$ and $q \leq r$, then $p \leq r$, so the composite $p \to r$ exists and is unique. Associativity holds automatically since there is at most one arrow between any two objects. Lastly, each $p \in P$ has an identity arrow $p \to p$ that corresponds to the reflexivity property of partial orders. Therefore, every poset defines a category. □

Example 4.1.4. Let us consider a simple functional programming language with the following types:

$$T ::= \mathrm{Nat} \mid \mathrm{Bool} \mid T \times T \mid T \to T$$

The type system is closed under product and function type constructors. A valid compound type is $(\mathrm{Nat} \times \mathrm{Bool}) \to \mathrm{Nat}$. A categorical interpretation of this scenario reveals types are objects and functions morphisms.

- Identity $\mathrm{id}_A : A \to A$ can be defined as $\mathrm{id}_A = \lambda x{:}A.\, x$, which is well-typed. Identity laws are verified trivially: $\mathrm{id}_B \circ f = f = f \circ \mathrm{id}_A$.

- Compositionality is defined as sequential application of functions that share domain and codomain through $f : A \to B$ and $g : B \to C$ like $g \circ f : A \to C$ by $g \circ f = \lambda x{:}A.\, g(f(x))$, which is well-typed by function application.

- Composition is associative due to β-equivalence in lambda calculus:

$$h \circ (g \circ f) = \lambda x{:}A.h(g(f(x))) = (h \circ g) \circ f$$

Therefore, the language satisfies all axioms of a category.

Example 4.1.5. **Hask** is often described as a category where the objects are Haskell types such as `Int`, `Bool`, `String`. The morphisms are Haskell functions between those types such as `f::Int->Bool`. For every type `a`, there is an identity morphism `id::a->a` that maps each value to itself. Morphisms can be composed using `(.)::(b->c)->(a->b)->a->c`, meaning the output of one function becomes the input of another. This composition is associative, and the identity morphisms act as neutral elements for composition.

Strictly speaking, Haskell types admit bottom values denoting non-termination or runtime errors, which means that most ordinary Haskell functions are not truly total. This can show up in definitions like:

```
loop :: Int
loop = loop
```

Because of this, **Hask** is not a category in the strict mathematical sense. Nonetheless, treating types as objects, functions as morphisms, `id` as identity, and `(.)` as composition provides a useful and widely accepted categorical model, at least as a first approximation.

Category	Description
Set	Sets as objects, functions as morphisms.
Grp	Groups and group homomorphisms.
Ring	Rings and ring homomorphisms.
$\mathbf{Mod}_R$	Modules over a ring R and R-linear maps.
$\mathbf{Vect}_k$	Vector spaces over field k and linear maps.
Top	Topological spaces and continuous functions.
Man	Smooth manifolds with smooth maps.
Hilb	Hilbert spaces and bounded linear maps.

Table 4.1: Key mathematical categories.

Category	Description
Mon	Monoids and monoid homomorphisms.
CPO	Complete partial orders and continuous maps.
Dom	Domains as posets with Scott-continuous maps.
Rel	Sets and binary relations as morphisms.
Pfn	Sets and partial functions as morphisms.
Par	Sets and partial maps (generalized morphisms).
$\mathbf{Kl}(T)$	Kleisli category of a monad T on a base category.
$\mathbf{Endo}(\mathbf{Set})$	Endofunctors on the category of sets.

Table 4.2: Key categories in computer science.

Duality

The categories we have built up until now, have been built by matching structures to the definition of category. However, if we look closely at any category we find that by reversing every morphism $f : A \to B$ into $f^{op} : B \to A$, we find ourselves with a newly created category in our hands. That is if we also reverse composition of morphisms.

Definition 4.1.4. For a category $\mathscr{C}$, the opposite or dual category $\mathscr{C}^{op}$ is defined by:

- $\mathrm{Ob}(\mathscr{C}^{op}) = \mathrm{Ob}(\mathscr{C})$
- For $A, B \in \mathrm{Ob}(\mathscr{C})$, $\mathscr{C}^{op}(A, B) = \mathscr{C}(B, A)$
- Composition is reversed: if $f : A \to B$ and $g : B \to C$ in $\mathscr{C}$, then $f^{op} \circ g^{op} = (g \circ f)^{op}$

This reversal allows any property or construction to be mirrored.

4.2 Monos, Epis and Isos

Monomorphisms, epimorphisms, and isomorphisms arise in category theory as abstract counterparts to injective, surjective, and bijective maps, but their definitions rely on morphisms instead of on elements.

Definition 4.2.1. Let $g_1, g_2 : X \to A$ be morphisms in a category $\mathscr{C}$. A morphism $f : A \to B$ is monic if

$$f \circ g_1 = f \circ g_2 \implies g_1 = g_2 \qquad \begin{tikzcd} X \arrow[r, shift left, "g_1"] \arrow[r, shift right, "g_2"'] & A \arrow[r, "f"] & B \end{tikzcd}$$

In other words, f is left-cancellable.

Definition 4.2.2. Let $h_1, h_2 : B \to Y$ be morphisms in a category $\mathscr{C}$. A morphism $f : A \to B$ is epic if

$$h_1 \circ f = h_2 \circ f \implies h_1 = h_2 \qquad \begin{tikzcd} A \arrow[r, "f"] & B \arrow[r, shift left, "h_1"] \arrow[r, shift right, "h_2"'] & Y \end{tikzcd}$$

In other words, f is right-cancellable.

Note 17. Monomorphisims generalize the notion of injectivity, while epimorphisms are the generalization of surjectivity.

Proposition 4.2.1. Given $\mathscr{C}$, monomorphisms in $\mathscr{C}$ are dual to epimorphisms in $\mathscr{C}^{op}$.

Proof. Let $\mathscr{C}$ be a category and $g_1, g_2 : X \to A, f : A \to B$, the dual category $\mathscr{C}^{op}$ has $g_1^{op}, g_2^{op} : A \to X, f^{op} : B \to A$ and so:

$$f \circ g_1 = f \circ g_2 \implies g_1 = g_2$$

in the dual category becomes

$$g_1^{op} \circ f^{op} = g_2^{op} \circ f^{op} \implies g_1^{op} = g_2^{op}$$

which is exactly the definition of epi in $\mathscr{C}$. It is equally easy to check that the converse is also true. □

Definition 4.2.3. An object A is isomorphic to B in a category $\mathscr{C}$ if there exists a morphism $f : A \to B$ such that there exists an inverse morphism $g : B \to A$ with

$$g \circ f = \mathrm{id}_A \quad \text{and} \quad f \circ g = \mathrm{id}_B.$$

Denoted $A \cong B$.

Proposition 4.2.2. Let $\mathscr{C}$ be a category and $f : A \to B$ a morphism in $\mathscr{C}$. If f is an isomorphism in $\mathscr{C}$, then f is also an isomorphism in the opposite category $\mathscr{C}^{\mathrm{op}}$.

Proof. Since f is an isomorphism in $\mathscr{C}$, there exists a morphism $g : B \to A$ such that

$$g \circ f = \mathrm{id}_A \quad \text{and} \quad f \circ g = \mathrm{id}_B.$$

In $\mathscr{C}^{\mathrm{op}}$, the morphisms f and g are reversed: $f^{\mathrm{op}} : B \to A$ and $g^{\mathrm{op}} : A \to B$. The compositions in $\mathscr{C}^{\mathrm{op}}$ satisfy

$$f^{\mathrm{op}} \circ g^{\mathrm{op}} = (g \circ f)^{\mathrm{op}} = \mathrm{id}_A, \quad g^{\mathrm{op}} \circ f^{\mathrm{op}} = (f \circ g)^{\mathrm{op}} = \mathrm{id}_B.$$

Hence f^{op} is an isomorphism in $\mathscr{C}^{\mathrm{op}}$. □

Isomorphisms are fundamental because they formalize the idea of structural equivalence between objects. As explained, Category Theory is mostly concerned with morphisms, which is why, isomorphisms play an important role. Many of the constructions we will introduce, such as products, coproducts, are unique only up to isomorphism, meaning, that they are unable to tell isomorphic objects apart from one another. This is one of the keys to the level of abstraction that Category Theory provides, only reaching up to isomorphism.

4.3 Functors

Category Theory takes itself very seriously. Hence, it is natural to consider functors, that are themselves morphisms in the category of categories. In this way, category theory can study itself. From this perspective, the definition of a functor arises naturally as a pair of functions that map both objects and arrows from one category to another while preserving identities and composition.

Definition 4.3.1. Let $\mathscr{C}$ and $\mathscr{D}$ be categories. A functor $F : \mathscr{C} \to \mathscr{D}$ consists of:

- A function that maps objects from $\mathscr{C}$ to $\mathscr{D}$:

$$\begin{aligned} F : \mathrm{Ob}(\mathscr{C}) &\longrightarrow \mathrm{Ob}(\mathscr{D}) \\ X &\longmapsto F(X) \end{aligned}$$

- A function that maps arrows in $\mathscr{C}$ to $\mathscr{D}$:

$$\begin{aligned} F : \mathrm{Hom}_{\mathscr{C}}(X, Y) &\longrightarrow \mathrm{Hom}_{\mathscr{D}}(F(X), F(Y)) \\ f &\longmapsto F(f) \end{aligned}$$

such that the properties of a category are kept across the transformation:

- For every object $X \in \mathscr{C}$, $F(\mathrm{id}_X) = \mathrm{id}_{F(X)}$.
- For all composable morphisms $f : X \to Y$ and $g : Y \to Z$ in $\mathscr{C}$,

$$F(g \circ f) = F(g) \circ F(f)$$

The following diagram elegantly summarizes the latter definition:

$$\begin{tikzcd} X \arrow[r, "f"] \arrow[d, mapsto, "F"'] & Y \arrow[d, mapsto, "F"] \\ F(X) \arrow[r, "F(f)"'] & F(Y) \end{tikzcd}$$

Example 4.3.1. Haskell captures the notion of functor using the following type class:

```
class Functor f where
    fmap :: (a -> b) -> f a -> f b
    (<$) :: a -> f b -> f a
    (<$) = fmap . const
```

There exist several instances for this class, among which we find:

```
instance Functor [] where
    fmap = map
```

That allows us to perform mapping operations on lists like:

```
ghci> fmap (*2) [1,2,3]
[2,4,6]
```

Example 4.3.2. A reformulation from example 2.1.2 in Pierce 1991 that defines a functor called List from the category **Set** → **Set** to illustrate functors from a programming perspective.

- For every set S, we define a new set:

$$\text{List}(S) := \{L \mid L \text{ is a finite list of elements of } S\}$$

  Where the order of elements matters, and duplicates are allowed.

- For every function $f : S \to S'$, we define a new function

$$\text{List}(f) : \text{List}(S) \to \text{List}(S')$$
$$\text{List}(f)([s_1, s_2, \ldots, s_n]) := [f(s_1), f(s_2), \ldots, f(s_n)].$$

  So, say for instance we have $S = \{1, 2\}$ and $f(s) = 2 \cdot s$. Then,

$$S' = \{2, 4\}, \text{List}(\{1, 2\}) = \{[], [1], [2], [1, 1], [1, 2], [2, 1], [2, 2], [1, 1, 1], \ldots\}$$

  and

$$\text{List}(f)(\{2, 4\}) = \{[], [2], [4], [2, 2], [2, 4], [4, 2], [4, 4], [2, 2, 2], \ldots\}$$

To confirm that List is a functor, it must satisfy the following two laws:

- Identity: For every set S and every list $L \in \text{List}(S)$: $\text{List}(\text{id}_S)(L) = L$.
- Composition: For any two functions $f : S \to S'$ and $g : S' \to S''$, and any list $L \in \text{List}(S)$: $\text{List}(g \circ f)(L) = (\text{List}(g) \circ \text{List}(f))(L)$.

Note 18. $(\text{List}(S), ++, [])$ is a monoid, where $++$ is associative list concatenation and $[]$ is the identity, satisfying $[] ++ L = L = L ++ []$.

We have so far used directed graphs to informally represent categories, now, we leverage functors to introduce a purpose-specific form of directed graph tailored to our setting. We refer to this fine-tuned graphs as diagrams, we emphasize commutative diagrams, those where all paths from any given object to any other are equivalent.

Definition 4.3.2. A diagram in category theory is a functor $F : D \to \mathscr{C}$ where D is the indexing category, that describes the diagram's shape, and $\mathscr{C}$ is the target category. Each object of D is mapped to an object of $\mathscr{C}$, and each morphism of D is mapped to a morphism of $\mathscr{C}$, preserving composition and identities.

Example 4.3.3. Let W, X, Y, Z be objects in a category $\mathscr{C}$ with morphisms

$$f : W \to X, \quad g : W \to Y, \quad h : X \to Z, \quad k : Y \to Z, \quad m : W \to Z.$$

The following diagram illustrates these morphisms:

$$\begin{tikzcd}
W \arrow[r, "f"] \arrow[d, "g"'] \arrow[dr, dashed, "m"] & X \arrow[d, "h"] \\
Y \arrow[r, "k"'] & Z
\end{tikzcd}$$

This diagram commutes if

$$m = h \circ f = k \circ g,$$

so all three paths from W to Z are equal. For a diagram to commute, this property should hold for any pair of objects.

Example 4.3.4. Consider the categories of groups **Grp** and sets **Set**. We define the functor $U : \textbf{Grp} \longrightarrow \textbf{Set}$:

- On objects: for each group $G \in \textbf{Grp}$, set $U(G)$ to be the set of elements of G.
- On morphisms: for each group homomorphism $f : G \to H$, we define $U(f) = f$, now considered as a function $U(G) \to U(H)$ between sets.

This mapping preserves identities and composition. Diagrammatically, we can represent the action of U on a morphism $f : G \to H$ as:

$$\begin{tikzcd}
G \arrow[r, "f"] \arrow[d, mapsto, "U"'] & H \arrow[d, mapsto, "U"] \\
U(G) \arrow[r, "U(f)"'] & U(H)
\end{tikzcd}$$

Since this functor "forgets" the structure its domain category we refer to it as forgetful functor.

Contravariant Functors

In category theory, the notion of a functor is introduced as a structure-preserving map between categories. When first defined, a functor is understood to preserve both the direction of morphisms and the way in which they compose. Since functors preserve the direction of arrows, they are usually called covariant functors. This terminology only becomes relevant once one contrasts them with contravariant functors, which reverse the direction of arrows.

Definition 4.3.3. Let $\mathscr{C}$ and $\mathscr{D}$ be categories. A contravariant functor $F : \mathscr{C} \to \mathscr{D}$ consists of:

- A function that maps objects from $\mathscr{C}$ to $\mathscr{D}$, same as 4.3.1:
- A function that maps arrows in $\mathscr{C}$ to arrows in $\mathscr{D}$, but reverses their direction:

$$\begin{aligned} F : \mathrm{Hom}_{\mathscr{C}}(X, Y) &\longrightarrow \underline{\mathrm{Hom}_{\mathscr{D}}(F(Y), F(X))} \\ f &\longmapsto F(f) \end{aligned}$$

such that identity and composition are preserved, same as 4.3.1. The following diagram summarizes how a contravariant functor reverses arrows:

$$\begin{tikzcd} X \arrow[r, "f"] \arrow[d, mapsto, "F"'] & Y \arrow[d, mapsto, "F"] \\ F(X) & F(Y) \arrow[l, "F(f)"'] \end{tikzcd}$$

The reason for distinguishing between covariant and contravariant functors can be clarified through duality. For any category $\mathscr{D}$, its opposite category $\mathscr{D}^{\mathrm{op}}$ has the same objects as $\mathscr{D}$, but with all morphisms reversed, such that: $\mathrm{Hom}_{\mathscr{D}^{\mathrm{op}}}(A, B) = \mathrm{Hom}_{\mathscr{D}}(B, A)$.

Proposition 4.3.1. Every covariant functor $F : \mathscr{C} \to \mathscr{D}$ can equally well be regarded as a contravariant functor $F : \mathscr{C} \to \mathscr{D}^{\mathrm{op}}$.

Proof. Suppose $F : \mathscr{C} \to \mathscr{D}$ is a covariant functor. Define a functor $F' : \mathscr{C} \to \mathscr{D}^{\mathrm{op}}$ by:

- On objects: $F'(A) := F(A)$ for all $A \in \mathscr{C}$.
- On morphisms: for $f : A \to B$ in $\mathscr{C}$, set $F'(f) := F(f) \in \mathrm{Hom}_{\mathscr{D}}(F(A), F(B))$. But in $\mathscr{D}^{\mathrm{op}}$ we have $\mathrm{Hom}_{\mathscr{D}}(F(A), F(B)) = \mathrm{Hom}_{\mathscr{D}^{\mathrm{op}}}(F(A), F(B))$, so $F'(f)$ is indeed a morphism in $\mathscr{D}^{\mathrm{op}}$.

Functoriality of F' follows immediately from functoriality of F, since identities and composition are preserved in $\mathscr{D}^{\mathrm{op}}$ exactly as in $\mathscr{D}$, but with reversed notation. Hence F' is a contravariant functor into $\mathscr{D}^{\mathrm{op}}$. □

Remark 26. Thus, what we call a *"covariant functor"* is just a functor $\mathscr{C} \to \mathscr{D}$, while a *"contravariant functor"* from $\mathscr{C}$ to $\mathscr{D}$ is a functor $\mathscr{C} \to \mathscr{D}^{\mathrm{op}}$. In this sense, the terminology covariant vs. contravariant simply reflects whether the functor sends morphisms into the target category itself or into its dual.

4.4 Natural Transformations

To define natural transformations we will apply the same reasoning scheme as we did with functors. We had categories and we defined functors as morphisms between categories, we do the same with natural transformations, that are nothing but morphisms from functors to functors. Suppose we have two functors $F, G : \mathscr{C} \to \mathscr{D}$. These represent two distinct ways of translating $\mathscr{C}$ into $\mathscr{D}$. A natural transformation between F and G then provides a systematic way to relate their outputs.

Definition 4.4.1. Let $\mathscr{C}$ and $\mathscr{D}$ be categories, and let $F, G : \mathscr{C} \to \mathscr{D}$ be functors. A natural transformation $\alpha : F \Rightarrow G$ between them:

$$\begin{tikzcd} \mathscr{C} \arrow[r, bend left=60, "F"] \arrow[r, bend right=60, "G"'] \arrow[r, phantom, "\Downarrow\alpha"] & \mathscr{D} \end{tikzcd}$$

is an assignment to every object $X \in \mathrm{Ob}(\mathscr{C})$ of a morphism $\alpha_X : F(X) \to G(X)$ in $\mathscr{D}$ called the component of α at X, such that for every morphism $f : X \to Y$ in $\mathscr{C}$, the following diagram commutes:

$$\begin{tikzcd} F(X) \arrow[r, "\alpha_X"] \arrow[d, "F(f)"'] & G(X) \arrow[d, "G(f)"] \\ F(Y) \arrow[r, "\alpha_Y"] & G(Y) \end{tikzcd}$$

An example from the `ghci` prompt that shows `reverse` is natural and preserves commutativity:

```
ghci> (reverse . (fmap toUpper)) "abc"
"CBA"
ghci> ((fmap toUpper) . reverse) "abc"
"CBA"
```

Definition 4.4.2. An endofunctor on a category $\mathscr{C}$ is a functor $F : \mathscr{C} \to \mathscr{C}$ that maps objects and morphisms of $\mathscr{C}$ to objects and morphisms of the same category, preserving identities and composition.

Example 4.4.1. A custom instantiation of functor similar to the previous example:

```
data BinTree a
    = Empty
    | Node a (BinTree a) (BinTree a)
    deriving (Show, Eq)

instance Functor BinTree where
    fmap _ Empty = Empty
    fmap f (Node x l r) = Node (f x) (fmap f l) (fmap f r)

reversePairs :: BinTree a -> BinTree a
reversePairs Empty = Empty
reversePairs (Node x l r) = Node x (reversePairs r) (reversePairs l)

testNaturality :: (Eq b) => (a -> b) -> BinTree a -> Bool
testNaturality f t =
    (reversePairs . fmap f) t == (fmap f . reversePairs) t
```

We now can call the `testNaturality` function to assert whether the `f` we provide makes the diagram commute.

```
ghci> f n = 2 * n
ghci> testNaturality f t
True
```

Note 19. From this example we extract a design pattern. For a given data structure whether a graph, a tree or otherwise, an operation that modifies structure can be considered a natural transform whenever we only deal with functions that don't modify structure.

In this case, any function that just operates on the keys will be a valid operation since it does not alter structure. Note how `testNaturality` just test naturality on a given argument. To support the claim that `reversePairs` it is in fact a natural transformation we proceed using structural induction.

Theorem 4.4.1. Let $\mathtt{RP}$, $\mathtt{No}$ and $\mathtt{Em}$ represent `reversePairs`, `Node` and `Empty` respectively. The function $\mathtt{RP} : \mathtt{BinTree} \to \mathtt{BinTree}$ defined by $\mathtt{RP}(\mathtt{Em}) = \mathtt{Em}$, $\mathtt{RP}(\mathtt{No}\ x\ l\ r) = \mathtt{No}\ x\ (\mathtt{RP}(r))\ (\mathtt{RP}(l))$ is a natural transformation. That is, for all functions $f : A \to B$ and trees $t : \mathtt{BinTree}\ A$, $\mathtt{fmap}\ f\ (\mathtt{RP}(t)) = \mathtt{RP}(\mathtt{fmap}\ f\ t)$.

Proof. We prove by structural induction on t.

- Base case: $t = \mathtt{Em}$. Then $\mathtt{fmap}\ f\ (\mathtt{RP}(\mathtt{Em})) = \mathtt{fmap}\ f\ (\mathtt{Em}) = \mathtt{Em} = \mathtt{RP}(\mathtt{Em}) = \mathtt{RP}(\mathtt{fmap}\ f\ \mathtt{Em})$ and the naturality condition holds.
- Inductive step: assume the naturality condition holds for subtrees l and r of depth $\leq n$. Let $t = \mathtt{No}\ x\ l\ r$. Then $\mathtt{RP}(t) = \mathtt{No}\ x\ (\mathtt{RP}(r))\ (\mathtt{RP}(l))$ and

$$\mathtt{fmap}\ f\ (\mathtt{RP}(t)) = \mathtt{fmap}\ f\ (\mathtt{No}\ x\ (\mathtt{RP}(r))\ (\mathtt{RP}(l))) = \mathtt{No}\ (f x)\ (\mathtt{fmap}\ f\ (\mathtt{RP}(r)))\ (\mathtt{fmap}\ f\ (\mathtt{RP}(l)))$$

By the inductive hypothesis, $\texttt{fmap}\ f\ (\texttt{RP}(l)) = \texttt{RP}(\texttt{fmap}\ f\ l), \texttt{fmap}\ f\ (\texttt{RP}(r)) = \texttt{RP}(\texttt{fmap}\ f\ r)$ so we can write:

$$\texttt{fmap}\ f\ (\texttt{RP}(t)) = \texttt{No}\ (f\,x)\ (\texttt{RP}(\texttt{fmap}\ f\ r))\ (\texttt{RP}(\texttt{fmap}\ f\ l)) = \texttt{RP}(\texttt{No}\ (f\,x)\ (\texttt{fmap}\ f\ l)\ (\texttt{fmap}\ f\ r)) = \texttt{RP}(\texttt{fmap}\ f\ t)$$

Hence the naturality condition holds for t.

By structural induction, RP is a natural transformation. □

Natural transformations, as well as functors can be composed as well, we study now the two ways there are of composing natural transformations:

Proposition 4.4.2. Given functors $F, G, H : \mathscr{C} \to \mathscr{D}$, natural transformations $\alpha : F \Rightarrow G$ and $\beta : G \Rightarrow H$ can be composed vertically:

$$\beta \circ \alpha : F \Rightarrow H$$

$$\begin{tikzcd}
\mathscr{C} \arrow[r, bend left=60, "F", "G"', "H"] \arrow[r, bend right=60] \arrow[r, phantom, bend left=20, "\alpha"] \arrow[r, phantom, bend right=20, "\beta"] & \mathscr{D}
\end{tikzcd}
\qquad
\begin{tikzcd}
F(X) \arrow[r, "\alpha_X"] \arrow[d, "F(f)"'] & G(X) \arrow[r, "\beta_X"] \arrow[d, "G(f)"'] & H(X) \arrow[d, "H(f)"] \\
F(Y) \arrow[r, "\alpha_Y"] & G(Y) \arrow[r, "\beta_Y"] & H(Y)
\end{tikzcd}$$

with components $(\beta \circ \alpha)_X = \beta_X \circ \alpha_X$ for each object $X \in \mathrm{Ob}(\mathscr{C})$.

Just as functors can be composed vertically, one after another, they can also be composed horizontally, a perspective that leads naturally to the study of natural transformations and 2-categorical structure.

Proposition 4.4.3. Given categories $\mathscr{C}, \mathscr{D}, \mathscr{E}$, functors $F, G : \mathscr{C} \to \mathscr{D}$ and $J, K : \mathscr{D} \to \mathscr{E}$, and natural transformations $\alpha : F \Rightarrow G$ and $\gamma : J \Rightarrow K$, we can define the horizontal composition:

$$\gamma * \alpha : J \circ F \Rightarrow K \circ G$$

$$\begin{tikzcd}
\mathscr{C} \arrow[r, bend left=60, "F", "G"'] \arrow[r, bend right=60] \arrow[r, phantom, bend left=20, "\alpha"] \arrow[r, phantom, bend right=20] & \mathscr{D} \arrow[r, bend left=60, "J", "K"'] \arrow[r, bend right=60] \arrow[r, phantom, bend left=20, "\gamma"] \arrow[r, phantom, bend right=20] & \mathscr{E}
\end{tikzcd}
\qquad
\begin{tikzcd}
J(F(X)) \arrow[r, "\gamma_{F(X)}"] \arrow[d, "J(\alpha_X)"'] & K(F(X)) \arrow[d, "K(\alpha_X)"] \\
J(G(X)) \arrow[r, "\gamma_{G(X)}"] & K(G(X))
\end{tikzcd}$$

with components $(\gamma * \alpha)_X = \gamma_{G(X)} \circ J(\alpha_X) = K(\alpha_X) \circ \gamma_{F(X)}$ for each $X \in \mathrm{Ob}(\mathscr{C})$.

Remark 27. The collection of all functors $F : \mathscr{C} \to \mathscr{D}$ together with natural transformations between them forms the functor category $\mathbf{Fun}(\mathscr{C}, \mathscr{D}) \in \mathrm{Ob}(\mathbf{Cat})$.

Example 4.4.2. This example builds the natural transform `reversePairs` from the composition of two variants of reverse pairs, one that reverses even depths, and another that does the converse on odd depths.

```
reverseDepth :: BinTree a -> Int -> BinTree a
reverseDepth Empty _ = Empty
reverseDepth (Node x l r) d
    | even d    = Node x (reverseDepth r (d+1)) (reverseDepth l (d+1))
    | otherwise = Node x (reverseDepth l (d+1)) (reverseDepth r (d+1))

reverseEven :: BinTree a -> BinTree a
reverseEven t = reverseDepth t 0

reverseOdd :: BinTree a -> BinTree a
reverseOdd t = reverseDepth t 1

horizReversePairs :: BinTree a -> BinTree a
horizReversePairs = reverseEven . reverseOdd

testHorizNaturality :: (Eq b) => (a -> b) -> BinTree a -> Bool
testHorizNaturality f t =
    (horizReversePairs . fmap f) t == (fmap f . horizReversePairs) t
```

Theorem 4.4.4. Let $\mathtt{RE} = \mathtt{reverseEven}$, $\mathtt{RO} = \mathtt{reverseOdd}$, and $\mathtt{RP} = \mathtt{reversePairs}$, and let $\mathtt{Em}$ and $\mathtt{No}$ denote $\mathtt{Empty}$ and $\mathtt{Node}$, respectively. Then for any binary tree t, $\mathtt{RP}(t) = (\mathtt{RO} * \mathtt{RE})(t)$.

Proof. We prove the statement by structural induction on t.

- Base case:
  - If $t = \mathtt{Em}$, then $\mathtt{RE}(\mathtt{Em}) = \mathtt{RO}(\mathtt{Em}) = \mathtt{RP}(\mathtt{Em}) = \mathtt{Em}$.
  - If $t = \mathtt{No}\ x\ \mathtt{Em}\ \mathtt{Em}$, then $\mathtt{RE}(t) = \mathtt{RO}(t) = \mathtt{RP}(t) = \mathtt{No}\ x\ \mathtt{Em}\ \mathtt{Em}$.
- Inductive step: Assume that for all subtrees of depth $\leq n$, the equality $\mathtt{RP}(s) = (\mathtt{RO} * \mathtt{RE})(s)$ holds. Let $t = \mathtt{No}\ x\ L\ R$ be a tree of depth $n+1$.
  - Applying $\mathtt{RE}$ gives $\mathtt{RE}(t) = \mathtt{No}\ x\ (\mathtt{RE}(R))\ (\mathtt{RE}(L))$.
  - Applying $\mathtt{RO}$ to $\mathtt{RE}(t)$ gives $\mathtt{RO}(\mathtt{RE}(t)) = \mathtt{No}\ x\ (\mathtt{RO}(\mathtt{RE}(R)))\ (\mathtt{RO}(\mathtt{RE}(L)))$.
  - By the inductive hypothesis, $\mathtt{RO}(\mathtt{RE}(L)) = \mathtt{RP}(L)$ and $\mathtt{RO}(\mathtt{RE}(R)) = \mathtt{RP}(R)$.
    Hence, $\mathtt{RO}(\mathtt{RE}(t)) = \mathtt{No}\ x\ (\mathtt{RP}(R))\ (\mathtt{RP}(L)) = \mathtt{RP}(t)$.

By structural induction, we conclude that $\mathtt{RP} = \mathtt{RO} * \mathtt{RE}$ for all binary trees t.

□

Corollary 4.4.5. $(\mathtt{RO} * \mathtt{RE})$ is a natural transformation.

Proof. Direct by 4.4.1 and 4.4.4. □

Remark 28. Compilers can use natural transforms to avoid performing unnecessary work and wasting precious resources. Take for instance the following `safeRoot` function that returns the element at the root of a tree if there is one:

```
safeRoot :: BinTree a -> Maybe (BinTree a)
safeRoot Empty = Nothing
safeRoot (Node x _ _) = Just (Node x Empty Empty)
```

Check, that by naturality, `(safeRoot . (fmap f))` achieves the same as `((fmap f) . safeRoot)`. So in the case that `f` is particularly expensive, we can apply `((fmap f) . safeRoot)` instead of `(safeRoot . (fmap f))`. The former more inexpensive because we first get rid of every element other than the root and then perform `f`, as opposed to performing `f` on every single node of the tree.

4.5 Universal Constructions

Universality is a central concept in category theory, capturing the idea that certain constructions are optimal or canonical in a precise sense. A universal construction ensures that any relevant morphism factors uniquely through a specific object and arrow. Universal arrows formalize this notion: they express a universal property from an object in one category to a functor into another. Moreover, by their very nature, universal arrows explain why universal properties are unique up to isomorphism.

Definition 4.5.1. Let $S : \mathscr{D} \to \mathscr{C}$ be a functor and let X be an object of $\mathscr{C}$. A universal arrow from X to S is a pair $\langle R, u \rangle$ consisting of an object R of $\mathscr{D}$ and a morphism $u : X \to S(R)$ in $\mathscr{C}$ such that for every object Y of $\mathscr{D}$ and every morphism $f : X \to S(Y)$ in $\mathscr{C}$, there exists a unique morphism $f' : R \to Y$ in $\mathscr{D}$ with $S(f') \circ u = f$.

This condition means that every arrow from X to $S(Y)$ factors uniquely through u, $S(f') \circ u = f$, as shown in the following commutative diagram:

$$\begin{tikzcd} X \arrow[r, "u"] \arrow[dr, "f"'] & S(R) \arrow[d, dashed, "S(f')"] & R \arrow[d, "f'"] \\ & S(Y) & Y \end{tikzcd}$$

Terminal & Initial Objects

Definition 4.5.2. An object T in a category $\mathscr{C}$ is terminal if for every object $X \in \mathrm{Ob}(\mathscr{C})$, there exists a unique morphism $!_X : X \to T$. This property characterizes T up to unique isomorphism.

Proposition 4.5.1. In the category of small categories, **Cat**, the point category 1 is terminal: for every small category $\mathscr{C}$, there exists a unique functor $!_{\mathscr{C}} : \mathscr{C} \to 1$. which sends every object of $\mathscr{C}$ to $*$ and every morphism to id_*.

Proof. Recall that the point category 1 has a single object $*$ and a single morphism $\mathrm{id}_* : * \to *$.

- Existence: Define $!_{\mathscr{C}}$ on objects by $!_{\mathscr{C}}(X) = *$ for every object $X \in \mathrm{Ob}(\mathscr{C})$, and on morphisms by $!_{\mathscr{C}}(f) = \mathrm{id}_*$ for every morphism f of $\mathscr{C}$. This assignment is well defined and preserves identities and composition:

$$!_{\mathscr{C}}(\mathrm{id}_X) = \mathrm{id}_* \quad \text{and} \quad !_{\mathscr{C}}(g \circ f) = \mathrm{id}_* = \mathrm{id}_* \circ \mathrm{id}_* = !_{\mathscr{C}}(g) \circ !_{\mathscr{C}}(f)$$

  so $!_{\mathscr{C}}$ is a functor $\mathscr{C} \to 1$.

- Uniqueness: If $F : \mathscr{C} \to 1$ is any functor, then for each object $X \in \mathscr{C}$ the object $F(X)$ must be an object of 1, hence $F(X) = *$. For each morphism $f : X \to Y$ in $\mathscr{C}$ the morphism $F(f) : * \to *$ must be a morphism of 1, and the only such morphism is id_*, so $F(f) = \mathrm{id}_*$. Thus F coincides with the above $!_{\mathscr{C}}$, proving uniqueness.

Therefore for every small category $\mathscr{C}$ there is a unique functor $\mathscr{C} \to 1$, i.e. 1 is a terminal object in **Cat**. □

Example 4.5.1. Let $\mathscr{C}$ be a category and let $F : 1 \to \mathscr{C}$ be the unique functor from the point category 1. For any object $C \in \mathscr{C}$, a terminal object T with the unique morphism $u : C \to T$ forms a universal arrow from C to F, because for any other morphism $f : C \to T$ (there is only one, by definition of terminal object), the identity morphism $\mathrm{id}_T : T \to T$ is the unique morphism making the diagram commute: $\mathrm{id}_T \circ u = f$. This shows that terminal objects are precisely universal arrows from the identity functor on 1 to $\mathscr{C}$.

Example 4.5.2. The following Haskell snippet provides a feasible definition for terminal object, where arguments of a given type `a`, are all collapsed to `()`.

```
terminate :: a -> ()
terminate _ = ()

ghci> terminate "I'm on death row :'("
()
```

Proposition 4.5.2. Let $\mathscr{C}$ be a category. Any two terminal objects of $\mathscr{C}$ are isomorphic. More precisely, if T and T' are terminal objects then there exists a unique isomorphism $i : T \to T'$, whose inverse is the unique map $T' \to T$.

Proof. Let T and T' be terminal objects of $\mathscr{C}$. By terminality of T' there is a unique morphism $i : T \to T'$, and by terminality of T there is a unique morphism $j : T' \to T$. Consider the composite $j \circ i : T \to T$. Since T is terminal, the identity id_T is the unique morphism $T \to T$, hence $j \circ i = \mathrm{id}_T$. Similarly, the composite $i \circ j : T' \to T'$ must equal $\mathrm{id}_{T'}$ because T' is terminal. Therefore i and j are mutual inverses, so i is an isomorphism with inverse j. Uniqueness: if $i' : T \to T'$ were any isomorphism, then by terminality of T' we must have $i' = i$, so the isomorphism is unique. □

Definition 4.5.3. By duality, initial objects are dual to terminal objects. An object I in a category $\mathscr{C}$ is initial if for every object $X \in \mathrm{Ob}(\mathscr{C})$, there exists a unique morphism $!_X : I \to X$ Dually, the point category 1 serves as an initial object in the opposite category $\mathbf{Cat}^{\mathrm{op}}$. Concretely, in **Cat** itself, there is also a unique functor $1 \to \mathscr{C}$ for any small category $\mathscr{C}$, sending $*$ to any chosen object of $\mathscr{C}$ and id_* to the identity morphism on that object.

Products and Coproducts

When we hear the word product, it is natural to think of the Cartesian product of sets: ordered pairs $\langle a, b\rangle$ with corresponding projections π_1, π_2. That is fine intuition in the category **Set**, but category theory is not about carrying over constructions element-by-element. Instead, it asks us to step back and characterize objects purely in terms of how morphisms interact with them.

So, when we speak of products, and later coproducts, in a category, we are not importing the set-theoretic construction. We are giving a new definition that says: the product is whatever object plays the right role with respect to morphisms. This shift in perspective is crucial: it frees us from thinking in terms of elements and lets the same idea apply in categories where "elements" do not make sense at all.

Definition 4.5.4. Let $\mathscr{C}$ be a category and let $A, B \in \mathrm{Ob}(\mathscr{C})$. A product of A and B is an object $A \times B$ together with morphisms $\pi_1 : A \times B \to A, \pi_2 : A \times B \to B$ such that for every object X with morphisms $f : X \to A$ and $g : X \to B$, there exists a unique morphism $\langle f, g\rangle : X \to A \times B$ making the diagram commute:

$$\begin{tikzcd}
 & X \arrow[dl, "f"'] \arrow[d, dashed, "\langle f,g\rangle"] \arrow[dr, "g"] & \\
A & A \times B \arrow[l, "\pi_1"] \arrow[r, "\pi_2"'] & B
\end{tikzcd}$$

$$\pi_1 \circ \langle f, g\rangle = f \quad \pi_2 \circ \langle f, g\rangle = g$$

Note 20. In these diagrams, the dashed arrow indicates a unique morphism guaranteed by the universal property of the product or coproduct. That is, for any mediating maps as in the diagram, there exists exactly one morphism making the diagram commute.

Proposition 4.5.3. From Imada 2019: In the category **Set**, the product of objects A and B is their Cartesian product $A \times B$.

Proof. Let X be a set and let $x_1 : X \to A$ and $x_2 : X \to B$. Consider the function

$$\begin{aligned} \langle x_1, x_2\rangle : X &\to A \times B \\ x &\mapsto (x_1(x), x_2(x)) \end{aligned}$$

By definition, we have $x_1 = \pi_1 \circ \langle x_1, x_2\rangle$ and $x_2 = \pi_2 \circ \langle x_1, x_2\rangle$. To show uniqueness of $\langle x_1, x_2\rangle$, suppose there is a morphism $f : X \to A \times B$ such that $x_1 = \pi_1 \circ f$ and $x_2 = \pi_2 \circ f$. Take $x \in X$ and denote

$$\langle x_1, x_2\rangle(x) = (a, b), \qquad f(x) = (a', b').$$

Then $a = \pi_1((a, b)) = \pi_1(f(x)) = x_1(x) = \pi_1(\langle x_1, x_2\rangle(x)) = \pi_1((a', b')) = a'$, and similarly $b = b'$. Hence $(a, b) = (a', b')$, so $f = \langle x_1, x_2\rangle$. Since (X, x_1, x_2) was arbitrary, it follows that $(A \times B, \pi_1, \pi_2)$ is the product of A and B in **Set**. □

Remark 29. From Imada 2019: Products are not unique. For example, in **Set** both $(A \times B, \pi_1, \pi_2)$ and $(B \times A, \pi_2, \pi_1)$ are products of A and B. However, there is a bijection $A \times B \to B \times A$, given by $(a, b) \mapsto (b, a)$. This means that products are unique up to isomorphism. In fact, in any category, every product of two objects is unique up to a unique isomorphism.

Definition 4.5.5. Let $\mathscr{C}$ be a category and let $A, B \in \mathrm{Ob}(\mathscr{C})$. A coproduct of A and B is an object $A + B$ together with morphisms $\iota_1 : A \to A + B, \iota_2 : B \to A + B$ such that for every object X with morphisms $f : A \to X$ and $g : B \to X$, there exists a unique morphism $[f, g] : A + B \to X$ making the diagram commute:

$$\begin{tikzcd}
A \arrow[r, "\iota_1"] \arrow[dr, "f"'] & A + B \arrow[d, dashed, "{[f,g]}"] & B \arrow[l, "\iota_2"'] \arrow[dl, "g"] \\
 & X &
\end{tikzcd}$$

$$[f, g] \circ \iota_1 = f \quad [f, g] \circ \iota_2 = g$$

Remark 30. Coproducts are dual to products.

Exponentials

Exponential objects model function types inside category theory. Given objects A and B, the exponential B^A models the type of functions from A to B. They come with an evaluation map $B^A \times A \to B$, which models function application.

Definition 4.5.6. Let $\mathscr{C}$ be a category with finite products. For objects $A, B \in \mathscr{C}$, an exponential object B^A is an object together with an evaluation morphism

$$\mathrm{ev}_{A,B} : B^A \times A \longrightarrow B$$

such that for every object $C \in \mathscr{C}$ and every morphism $f : C \times A \longrightarrow B$ there exists a unique morphism $\lambda f : C \longrightarrow B^A$ making the following diagram commute:

$$\begin{tikzcd} C \times A \arrow[rr, dashed, "\lambda f \times 1_A"] \arrow[dr, "f"'] & & B^A \times A \arrow[dl, "\mathrm{ev}_{A,B}"] \\ & B & \end{tikzcd}$$

That is, $\mathrm{ev}_{A,B} \circ (\lambda f \times 1_A) = f$.

Proposition 4.5.4. The exponential object B^A, if it exists, is unique up to unique isomorphism.

The definition above says that in a category with finite products, an exponential object B^A behaves like the space of functions $A \to B$. The evaluation morphism $\mathrm{ev}_{A,B} : B^A \times A \longrightarrow B$ expresses function application: given a function and an input, one can obtain an output in B. The universal property states that any morphism $f : C \times A \longrightarrow B$ can be uniquely factored through B^A by a morphism $\lambda f : C \longrightarrow B^A$ such that $f = \mathrm{ev}_{A,B} \circ (\lambda f \times 1_A)$.

Remark 31. This captures the idea of currying: turning a function of two arguments into a function that returns a function.

$$f = \lambda(a, b).\, g(a, b) \quad \rightsquigarrow \quad \lambda f = \lambda a.\, \lambda b.\, g(a, b)$$

We can obtain effectively equivalent behavior in Haskell using partial application:

```
mulBy :: Int -> (Int -> Int)
mulBy n = \m -> n * m
triple :: Int -> Int
triple = mulBy 3
```

Remark 32. The function `mulBy` takes an integer n and returns a new function that multiplies its argument by n. Defining `triple = mulBy 3` is an instance of partial application: we supply the first argument, leaving a function that still expects the second. Thus `triple 4` evaluates to 12.

Remark 33. From the type-theoretic perspective, B^A corresponds to the function type $A \to B$. The evaluation morphism is ordinary application, and the universal property is precisely the λ-abstraction rule: if a term of type B can be defined assuming a variable of type A, then one obtains a function of type $A \to B$.

The universal constructions we have studied so far—products, coproducts, terminal objects, and exponentials—are by no means the only ones of importance in category theory. Other central constructions include equalizers and coequalizers, pullbacks and pushouts, as well as more general limits and colimits. For a broader treatment of these notions, one can consult standard references in category theory, such as *Categories for the Working Mathematician* Lane 1971, *Category Theory for Computing Science* Barr and Wells 1998 and *Basic Category Theory for Computer Scientists* Pierce 1991.

4.6 Closed Cartesian Categories

A cartesian closed category is a category with three key pieces of structure. First, it has a terminal object, which plays the role of a unique trivial object that every other object admits a morphism to. Second, it has products, which capture the idea of combining objects together, with canonical ways of recovering each component. Finally, it has exponentials, which express that the process of mapping from one object to another can itself be represented as an object in the category. In other words, the collection of morphisms from one object to another is not just external data, but has an internal representation.

Definition 4.6.1. A category $\mathscr{C}$ is called cartesian closed or closed cartesian if:

- $\mathscr{C}$ has a terminal object 1 and all binary products $A \times B$.
- for every pair of objects $A, B \in \mathscr{C}$ there exists an object B^A, called the exponential together with an evaluation morphism $\mathrm{ev}_{A,B} : B^A \times A \longrightarrow B$ satisfying the following universal property: for every object C and every morphism $f : C \times A \longrightarrow B$ there is a unique morphism $\lambda f : C \longrightarrow B^A$ called the transpose or curried map such that

the triangle below commutes:

$$\begin{tikzcd} C \times A \arrow[r, "f"] \arrow[d, dashed, "\lambda f \times \mathrm{id}_A"'] & B \\ B^A \times A \arrow[ur, "\mathrm{ev}_{A,B}"'] & \end{tikzcd}$$

Thus, a CCC is a setting where not only objects and morphisms exist, but also the very process of mapping between objects is itself represented as an object of the category. In this way, CCCs resemble deduction systems: every morphism can be systematically built from basic constructions, just as every proof is assembled step by step from simpler proofs.

Cartesian closed categories provide a natural framework for viewing categories as deduction systems. They bring together products, exponentials, and the terminal object in a way that mirrors the structure of logic and type theory. In the next chapter, we will build further on this connection.

4.7 Adjunctions and The Yoneda Lemma

In this optional section we look at how how the universal constructions we have seen so far –products, coproducts, exponentials– are just different faces of the same idea: adjointness. You can think of an adjunction as a "best possible solution" to a certain problem: a product is the best way to combine things, a coproduct is the best way to branch, and exponentials are the best way to internalize functions. So instead of memorizing each universal property separately, you can see them all as special cases of this one guiding principleone functor "undoes" or "complements" another in the most universal way possible.

Remark 34. The following section is included to provide additional intuition and insight into category-theoretic concepts, such as adjunctions and universal constructions. It is not the main focus of this work, which is why it is kept short, formal, and does not contain detailed examples or extended explanations. It is meant to show the flexibility of categorical abstractions when it comes to defining new universal constructions. We also include the Yoneda lemma in an effort to show the tools that category theory brings to the table.

Adjunctions

Notice how this notion of "best possible solution", closely resembles the idea of universal arrow from 4.5.1. The universal constructions we covered all fall under the umbrella of adjoint functors.

Definition 4.7.1. An adjunction between two categories $\mathscr{C}$ and $\mathscr{D}$ consists of a pair of functors $F : \mathscr{C} \to \mathscr{D}, G : \mathscr{D} \to \mathscr{C}$ together with a natural isomorphism between hom-sets:

$$\mathrm{Hom}_{\mathscr{D}}(F(C), D) \cong \mathrm{Hom}_{\mathscr{C}}(C, G(D))$$

for all $C \in \mathscr{C}$ and $D \in \mathscr{D}$. We say that F is left adjoint to G, and G is right adjoint to F.

$$\begin{tikzcd} F(C) \arrow[r, "f"] & D \\ C \arrow[u, "F"] \arrow[r, dashed, "g"'] & G(D) \arrow[u, "G"'] \end{tikzcd}$$

The naturality condition means that these bijections assemble into natural isomorphisms of functors:

- For a fixed $c \in \mathscr{C}$
$$\mathscr{C}(F-, c) : \mathscr{D} \to \mathbf{Set}^{\mathrm{op}} \quad \cong \quad \mathscr{D}(-, Gc) : \mathscr{D} \to \mathbf{Set}^{\mathrm{op}}$$
- For a fixed $d \in \mathscr{D}$
$$\mathscr{C}(Fd, -) : \mathscr{C} \to \mathbf{Set} \quad \cong \quad \mathscr{D}(d, G-) : \mathscr{C} \to \mathbf{Set}$$

Note 21. The symbol "-" is a placeholder for a variable argument: e.g. $D(-, Gc)$ denotes the functor $X \mapsto D(X, Gc)$, and $D_C(Fd, -)$ denotes $Y \mapsto D_C(Fd, Y)$. Parentheses are usually omitted for clarity.

The functor F is called the left adjoint to G, while G is the right adjoint to F, and we write

$$F \dashv G.$$

Adjunctions can be viewed as a weak form of equivalence of categories: every equivalence induces an adjunction, although not every adjunction is an equivalence. In many cases, an adjunction can be upgraded to an equivalence by suitable natural modifications of the categories and functors involved.

Remark 35. Universal arrows and adjunctions are closely related: a universal arrow from an object $C \in \mathscr{C}$ to a functor $G : \mathscr{D} \to \mathscr{C}$ can be thought of as a best possible arrow from C into the image of G. If G admits a universal arrow for every object $C \in \mathscr{C}$, then these universal arrows assemble into a left adjoint F to G, giving the adjunction $F \dashv G$. Conversely, given an adjunction $F \dashv G$, the components of its unit $\eta_C : C \to G(F(C))$ are precisely universal arrows. In this sense, universal arrows are the local building blocks of adjunctions.

Using adjunctions we can formulate alternative definitions for the universal constructions we already covered. To illustrate, we give a concise definition for product:

Definition 4.7.2. Let $\mathscr{C}$ be a category. Define the diagonal functor $\Delta : \mathscr{C} \to \mathscr{C} \times \mathscr{C}, X \mapsto (X, X)$. Think of a matrix with coordinates x, y to understand why the functor is called "diagonal". A product in $\mathscr{C}$ is a functor

$$\times : \mathscr{C} \times \mathscr{C} \to \mathscr{C} \qquad (A, B) \mapsto A \times B,$$

which is right adjoint to Δ. That is, there exists a natural bijection

$$\mathrm{Hom}_{\mathscr{C}}(X, A \times B) \;\cong\; \mathrm{Hom}_{\mathscr{C}\times\mathscr{C}}(\Delta X, (A, B))$$

for all $X \in \mathscr{C}$ and $(A, B) \in \mathscr{C} \times \mathscr{C}$.

Remark 36. Unpacking the adjunction shows that giving a morphism $X \to A \times B$ is equivalent to giving a pair of morphisms $X \to A$ and $X \to B$. Thus, the product object $A \times B$ comes equipped with projections $\pi_A : A \times B \to A$ and $\pi_B : A \times B \to B$, which satisfy the universal property of products.

We will not redefine all of the already presented universal constructions, Lane 1971 presents other constructions like limits and colimits by means of adjunctions.

The Yoneda Lemma

In the 1950s, Yoneda observed a striking fact: one never needs to "touch" an object of a category directly to know it. All the information about the object is already encoded in how other objects map into it. The essence of Yoneda is this: an object is its pattern of relationships. The Yoneda lemma formalizes this intuition. To understand an object, one examines its interactions with all other objects via morphisms. Concretely, every object X is associated with a functor of points

$$\mathrm{Hom}(-, X) \quad \text{or dually} \quad \mathrm{Hom}(X, -)$$

which records how X receives or sends morphisms. This functor fully characterizes X. The key technical insight is that natural transformations from a representable functor $\mathrm{Hom}(-, X)$ to any other functor F correspond bijectively to elements of $F(X)$. Intuitively, a single "universal element" encodes the entire transformation. This is why the Yoneda lemma is so powerful: it turns abstract objects into concrete data about morphisms, and it reduces questions about naturality to the behavior of one element. In short, the Yoneda lemma builds the bridge between structure and behavior. It shows that categories are not defined by what their objects "are," but by how they relate.

Theorem 4.7.1. The Yoneda Lemma: Let $\mathscr{C}$ be a locally small category, let $X \in \mathrm{Ob}(\mathscr{C})$, and let $F : \mathscr{C} \to \mathbf{Set}$ be a functor. Then there is a natural bijection

$$\mathrm{Nat}\big(\mathrm{Hom}_{\mathscr{C}}(-, X), F\big) \;\cong\; F(X)$$

where $\mathrm{Hom}_{\mathscr{C}}(-, X) : \mathscr{C}^{op} \to \mathbf{Set}$ is the representable functor sending $A \mapsto \mathrm{Hom}_{\mathscr{C}}(A, X)$. This Nat in the Yoneda Lemma refers to natural transformations between functors. It denotes the set $\mathrm{Nat}(\mathrm{Hom}(C, -), F)$, of all natural transformations from the Hom-functor to F.

Remark 37. The bijection is natural in both X and F. Concretely, a natural transformation $\eta : \mathrm{Hom}_{\mathscr{C}}(-, X) \Rightarrow F$ is uniquely determined by the element $\eta_X(\mathrm{id}_X) \in F(X)$. Thus, all of η's data is encoded in the image of the identity morphism. This is the "universal element" perspective: knowing how X is represented inside F fixes the entire transformation.

Example 4.7.1. Let $\mathscr{C} = \mathbf{Set}$ and fix a set X. Consider the functor of points

$$\mathrm{Hom}_{\mathbf{Set}}(-, X) : \mathbf{Set}^{op} \to \mathbf{Set}, \quad A \mapsto \mathrm{Hom}_{\mathbf{Set}}(A, X).$$

Take another functor $F : \mathbf{Set}^{op} \to \mathbf{Set}$, for instance $F(A) = \mathscr{P}(A)$ (the power set of A). By Yoneda, natural transformations $\mathrm{Hom}_{\mathbf{Set}}(-, X) \Rightarrow \mathscr{P}(-)$ are in bijection with elements of $\mathscr{P}(X)$, i.e. subsets of X. Indeed, given $S \subseteq X$, one defines a natural transformation by sending each map $f : A \to X$ to the preimage $f^{-1}(S) \subseteq A$. Different choices of S give different natural transformations. Hence, the Yoneda lemma tells us that the family of all such natural transformations is nothing more than the collection of subsets of X.

The Yoneda lemma tells us that an object in a category is completely determined by the morphisms coming into it. Building on this idea, instead of studying a category $\mathscr{C}$ directly, we can study all functors from $\mathscr{C}$ into **Set**, the category of sets. This approach has several advantages:

- **Set** is a category we already understand very well.
- Each functor $\mathscr{C} \to \mathbf{Set}$ translates the abstract structure of $\mathscr{C}$ into something concrete, providing a "representation" of $\mathscr{C}$ in terms of sets and functions.
- The original category $\mathscr{C}$ embeds into this functor category via representable functors, but the functor category also contains new objects that did not exist in $\mathscr{C}$. Treating these new objects like ordinary objects often unifies and simplifies the theory.

In short, studying functors from $\mathscr{C}$ to **Set** lets us understand $\mathscr{C}$ indirectly using concrete structures, often revealing hidden relationships and simplifying the theory.

References & Resources The categorical content of this chapter is based on foundational works in category theory. Mac Lane Lane 1971 provides the formal structure of categories, functors, and natural transformations; Pierce Pierce 1991 and Barr and Wells Barr and Wells 1998 are used to connect Mac Lane's ideas with computer science. Content related to Lambek is also influenced by Jacobs 1999.

Chapter 5

Curry–Howard–Lambek: Categorical Semantics

Categorical semantics provides a formal, structural interpretation of type systems using the language of category theory. Having introduced typed λ-calculus and the Curry–Howard correspondence, we have seen how types correspond to propositions and terms to proofs. These systems can be viewed as deduction systems, where derivations are constructed according to inference rules, identities correspond to trivial proofs, and composition of derivations reflects sequential application of rules. This perspective emphasizes the compositional nature of proofs: complex derivations are built from simpler ones, and the structure of the system ensures correctness and consistency.

Categories provide a natural generalization of this viewpoint. A category can itself be interpreted as an abstract deduction system in which objects represent propositions or types, morphisms represent proofs or derivations, and composition of morphisms corresponds to chaining proofs, while identity morphisms correspond to reflexive or trivial derivations. Categorical constructions naturally mirror logical operations and type constructors: products correspond to conjunctions, coproducts correspond to disjunctions, and exponentials correspond to implication or function types. In this way, a category encodes the structural relationships between types and proofs, providing a compositional framework that abstracts and extends the reasoning we observed in typed λ-calculus.

This perspective also suggests a natural pathway for further generalization. Just as Curry–Howard relates types and proofs, one can extend the correspondence to Curry–Howard–Lambek, interpreting typed λ-calculi fully within categorical structures. This extension allows reasoning not only about individual proofs and terms but about entire type systems and their semantic behavior, opening the door to categorical models of richer logical systems and functional programming languages.

5.1 Categories as Deduction Systems

Closed Cartesian Categories can be understood as abstract deduction systems: objects correspond to formulas, morphisms to proofs, composition to chaining deductions, and identities to trivial self-deductions. The axioms of category theory then capture the structural rules that govern reasoning itself.

- Objects correspond to propositions or types.
- Morphisms correspond to proofs or derivations: a morphism $f : A \to B$ can be seen as a proof that A entails B.
- Composition of morphisms corresponds to composition of proofs: if $f : A \to B$ and $g : B \to C$, then $g \circ f : A \to C$ represents the derived proof.
- Identity morphisms correspond to trivial or reflexive proofs (A implies A).

Informally:

$$\text{Objects} \leftrightarrow \text{Types} \leftrightarrow \text{Propositions}$$

$$\text{Arrows} \leftrightarrow \text{Terms} \leftrightarrow \text{Proofs}$$

$$\text{Category structure} \leftrightarrow \text{Type System} \leftrightarrow \text{Deduction system}$$

referred to as the Curry–Howard–Lambek correspondence. In this view, we can look at the typing extensions presented and come up with the following:

- Products model conjunction: $A \times B$ represents $A \wedge B$, with projections corresponding to elimination rules.
- Coproducts model disjunction: $A + B$ represents $A \vee B$, with injections corresponding to introduction rules.
- Exponentials B^A model implication: a morphism $f : C \times A \to B$ corresponds to a proof of $A \to B$ under context C.

Proof that $\mathscr{C}(\lambda)$ is Cartesian Closed

In this section, we conclude our development by showing that the category $\mathscr{C}(\lambda)$ is cartesian closed. We establish the existence of a terminal object and binary products, ensuring that typed λ-calculus is cartesian. We then prove the existence of exponentials, verifying that function spaces can be represented internally. Together, these results show that typed λ-calculus possesses the full structure of a cartesian closed category.

Definition 5.1.1. $\mathscr{C}(\lambda)$ is defined as:

- Objects are types of the simply typed λ-calculus: $\mathrm{Ob}(\mathscr{C}(\lambda)) = \mathscr{T}^{\times}$.
- For types A, B, morphisms $A \to B$ are closed terms $c \vdash A \to B$ modulo $\beta\eta$-equivalence. We can agree that morphisms between types are defined depending on whether terms that inhabit them are $\beta\eta$-equivalent to each other.
- We define identity for every type A as $\mathrm{id}_A := \lambda x : A.x$, such that, for every term y of type A we have $(\lambda x : A.x)\, y \rightsquigarrow_{\beta\eta} y \vdash A$.
- Given $b : A \to B$ and $c : B \to C$, we can define composition as: $c \circ b := \lambda x.\, c(bx)$.

Proposition 5.1.1. $\mathscr{C}(\lambda)$ is a well-defined category.

Proof of well-definedness. To be certain this is a category, compositionality must hold, as well as ssociativity laws and identity laws.

- Thanks to typing we can compose functions as long as the output type of the first matches the input type of the second. However, we need to check that no matter what representative we choose for b and c we still get the same result. Suppose $b \equiv_{\beta\eta} b'$ and $c \equiv_{\beta\eta} c'$, then $\lambda x.\, c(bx) \equiv_{\beta\eta} \lambda x.\, c'(b'x)$ also holds. This is due to the fact that $\beta\eta$-equivalence is preserved under contexts, in particular abstraction. Hence, composition is well defined.
- To check for associativity: Let $a : C \to D$, $b : B \to C$, and $c : A \to B$:

$$\begin{aligned}(a \circ b) \circ c &= \lambda x : A.\, (a \circ b)(c\, x) \\ &= \lambda x : A.\, (\lambda y : B.\, a(b\, y))(c\, x) \\ &\rightsquigarrow_{\beta} \lambda x : A.\, a(b(c\, x)) \quad \vdash A \to D.\end{aligned} \qquad \begin{aligned}a \circ (b \circ c) &= \lambda x : A.\, a((b \circ c)\, x) \\ &= \lambda x : A.\, a((\lambda y : A.\, b(c\, y))\, x) \\ &\rightsquigarrow_{\beta} \lambda x : A.\, a(b(c\, x)) \quad \vdash A \to D.\end{aligned}$$

  We conclude $(a \circ b) \circ c \equiv_{\beta\eta} a \circ (b \circ c)$. Thus, composition is associative.
- To check that identity rules are satisfied, let $b : A \to B$.

$$\begin{aligned}\mathrm{id}_B \circ b &= \lambda x : A.\, (\lambda y : B.\, y)(b\, x) \\ &\rightsquigarrow_{\beta} \lambda x : A.\, b\, x \\ &\rightsquigarrow_{\beta} b \quad \vdash A \to B.\end{aligned} \qquad \begin{aligned}b \circ \mathrm{id}_A &= \lambda x : A.\, b((\lambda y : A.\, y)\, x) \\ &\rightsquigarrow_{\beta} \lambda x : A.\, b\, x \\ &\rightsquigarrow_{\beta} b \quad \vdash A \to B.\end{aligned}$$

  Hence both left and right identity laws hold.

Therefore, $\mathscr{C}(\lambda)$ is a category. □

Proposition 5.1.2. $\mathscr{C}(\lambda)$ forms a Cartesian Closed Category.

Proof. We know from 5.1.1 that $\mathscr{C}$ is a category. Now, we have to find the representatives for terminal, product and exponential and verify their respective properties.

- Take the unit type 1. Define $!_A : A \to 1$ by $!_A := \lambda x.\, *$ for every type A. Where $* : 1$. If $f : A \to 1$ is any morphism, then $f\, a : 1$ for each a, hence $f\, a \equiv_{\beta\eta} *$. Thus $f \equiv_{\beta\eta} !_A$. So 1 is terminal. Note the grammar for types only defines 1 as terminal type, we know from our categorical background that all terminal objects are isomorphic to each other, so from a categorical perspecitive it would not make sense to add more terminal objects since results would not be altered.

- The type $A \times B$ together with the projections π_1 and π_2, forms a categorical product. Given $f : C \to A$ and $g : C \to B$, define the pairing like this: $\langle f, g\rangle := \lambda x : C.\, \langle fx,\ gx\rangle : C \to A \times B$.
  - We check that the projection laws hold:

$$\pi_1 \circ \langle f, g\rangle = \lambda x : C.\, \pi_1(\langle fx, gx\rangle) \rightsquigarrow_\beta \lambda x : C.\, fx \equiv_\eta f$$

$$\pi_2 \circ \langle f, g\rangle = \lambda x : C.\, \pi_2(\langle fx, gx\rangle) \rightsquigarrow_\beta \lambda x : C.\, gx \equiv_\eta g.$$

  - Then, to check for uniqueness we suppose that $h : C \to A \times B$ satisfies $\pi_1 \circ h \equiv_{\beta\eta} f$ and $\pi_2 \circ h \equiv_{\beta\eta} g$. Such that for each $c : C$ we have $hc \equiv_{\beta\eta} \langle \pi_1(hc), \pi_2(hc)\rangle \equiv_{\beta\eta} \langle fc,\ gc\rangle$, hence $h \equiv_{\beta\eta} \langle f, g\rangle$.

  Therefore $A \times B$ together with π_1, π_2 and $\langle f, g\rangle$ satisfies the universal property of a product and that the diagram 4.5.4 commutes in this instance.

- To find the exponential construction structure for A, B, take $A \Rightarrow B := A \to B$. We define the evaluation map

$$\mathrm{ev} : (A \to B) \times A \to B \qquad \mathrm{ev} := \lambda p.\, (\pi_1 p)(\pi_2 p)$$

  Given $h : C \times A \to B$, define its curry $\mathrm{curry}(h) := \lambda c.\ \lambda a.\ h\langle c, a\rangle$, then $(\mathrm{ev} \circ \langle \mathrm{curry}(h), \pi_2\rangle)p \rightsquigarrow_\beta hp$, so $\mathrm{ev} \circ \langle \mathrm{curry}(h), \pi_2\rangle \equiv_{\beta\eta} h$. Conversely, if $k : C \to (A \to B)$ satisfies $\mathrm{ev} \circ \langle k, \pi_2\rangle \equiv h$, then for any c, a:

$$(kc)\, a \equiv h\langle c, a\rangle$$

  so $kc \equiv \lambda a.\ h\langle c, a\rangle$, hence $k \equiv \mathrm{curry}(h)$. Thus $\mathrm{Hom}(C \times A, B) \cong \mathrm{Hom}(C, A \to B)$. Hence $A \to B$ is the exponential.

We have exhibited a terminal object, binary products, and exponentials satisfying the required universal properties. Therefore, to 5.1.1, we add that $\mathscr{C}(\lambda)$ is Cartesian Closed. □

The Curry–Howard–Lambek correspondence provides a unifying perspective between intuitionistic logic, typed λ-calculus, and category theory. Logical connectives, type formers, and categorical constructions relate to each other as summarized in Table 5.1.

Table 5.1: The Curry–Howard–Lambek Correspondence

Intuitionistic Logic	**Typed λ-Calculus**	**Category Theory / Lambek**
Proposition A	Type A	Object A
Proof of A	Term of type A	Morphism $f : 1 \to A$
Implication $A \to B$	Function type $A \to B$	Exponential object B^A
Conjunction $A \wedge B$	Product type $A \times B$	Product $A \times B$ and projections
$\top$	1	Terminal object 1
Proof normalization	β-reduction	Commutativity of diagrams

This basic correspondence can be extended further. For instance, when we add sum types to the typed λ-calculus, they correspond to disjunctions in logic and coproducts in category theory:

Table 5.2: Extension of the Curry–Howard–Lambek Correspondence with Sum Types

Intuitionistic Logic	**Typed λ-Calculus**	**Category Theory / Lambek**
Disjunction $A \vee B$	Sum type $A + B$	Coproduct $A + B$ and injections
Falsity $\bot$	Empty type 0	Initial object 0

Thus, we informally extended the Curry–Howard–Lambek triangle to incorporate additional logical and categorical structure via sum types and coproducts. This is further discussed in Jacobs 1999; Lane 1971; Girard 1989; Pierce 1991; Barr and Wells 1998.

5.2 Closing Remarks and Perspectives

This work has provided a comprehensive journey through the pillars of modern theoretical computer science: the untyped and typed λ-calculus, intuitionistic logic, and category theory, building up to the profound Curry–Howard–Lambek correspondence.

Closing Remarks

We began with the untyped λ-calculus, a minimal yet powerful model of computation. We explored its syntax, reduction semantics, covering: α, β, and η-equivalence. As well as key metatheoretic properties such as confluence, expressed through the Church–Rosser theorem and normalization. While the untyped calculus is Turing-complete, its lack of structure leads to undecidability of termination and the presence of non-normalizing terms like Ω. This motivated the introduction of typing disciplines.

In the typed λ-calculus, we examined both Church-style, explicitly annotated, and Curry-style , implicitly inferred systems. We extended the minimal simply typed calculus with product and sum types, enriching its expressive power while retaining desirable properties like strong normalization. The Church–Rosser property was revisited in this extended setting, noting that while β-reduction remains confluent, the addition of η-rules can break confluence with the presence of unit types. The proof of weak normalization via redex degrees illustrated how typing enforces structure and guarantees termination.

The Curry–Howard isomorphism then revealed a deep connection between computation and logic: types correspond to propositions, terms to proofs, and β-reduction to proof normalization. This propositions-as-types paradigm not only provides a computational interpretation of intuitionistic logic but also grounds programming in a rigorous logical framework. Through natural deduction and the Brouwer–Heyting–Kolmogorov interpretation, we saw how constructive proofs align with functional programs.

We then introduced category theory as a universal language for structure and composition. By abstracting away from internal details and focusing on morphisms and their interactions, category theory offers a unifying perspective across mathematics and computer science. We covered basic constructs—categories, functors, natural transformations—and key universal properties such as products, coproducts, exponentials, and the definition of cartesian closed categories. Examples drawn from programming made these abstract concepts tangible and illustrated their practical relevance.

Finally, we arrived at the Curry–Howard–Lambek correspondence, which integrates these three domains into a cohesive whole:

$$\text{Logic} \longleftrightarrow \text{Type Theory} \longleftrightarrow \text{Category Theory}.$$

We showed that the category of simply typed λ-terms is cartesian closed, thereby providing a categorical model of both logic and computation. This three-way correspondence not only estabishes similarities between these fields but also enables the transfer of techniques and insights across domains.

Perspectives

This work lays the groundwork for several advanced topics, where logic, type theory, and category theory continue to develop:

- Polymorphic types extend the simply typed setting by allowing terms to be parameterized over types, so that a single definition can be used uniformly across many different instances. In System F, also known as the polymorphic λ-calculus, this is expressed through universal quantification over type variables:

  $$T ::= \alpha \mid T \to T \mid \forall \alpha . T$$

  where α ranges over type variables. A familiar example is the polymorphic identity function,

  $$\Lambda \alpha . \lambda x : \alpha . x : \forall \alpha . \alpha \to \alpha$$

  which works for every type without modification. For any particular type A, applying this term yields the identity function $\lambda x : A . x : A \to A$, so every input is returned unchanged and its type remains unaltered. This makes it possible to define data structures and algorithms in a uniform way independently of their type. For example, to the map function has type:

  $$\texttt{map} : \forall \alpha\beta . (\alpha \to \beta) \to \texttt{List}(\alpha) \to \texttt{List}(\beta)$$

  that expresses the principle that functions between elements can be lifted to functions between lists. The key point is that these definitions do not depend on any particular type, but apply uniformly across all types, since we quantified α and β such that `map` works on every type.

  From a logical perspective, System F corresponds to second-order intuitionistic logic, where quantification ranges over types. This extension allows reasoning about entire families of types and proofs, making polymorphism a bridge between the simply typed world and more expressive systems such as dependent types.

From a categorical perspective, modeling polymorphism requires more than the cartesian closed categories that suffice for the simply typed λ-calculus. A type with type variables α, say $T(\alpha)$, is interpreted as a functor in the parameter α, meaning that T can be instantiated by any concrete type, and functions between types induce corresponding functions between the instantiated versions. For example, $T(\alpha) = \texttt{List}(\alpha)$ yields $\texttt{List}(\mathbb{N})$ when α is the type of natural numbers and $\texttt{List}(\texttt{Bool})$ when α is the type of booleans. Moreover, a function $f : A \to B$ gives rise to a function $\texttt{map}(f) : \texttt{List}(A) \to \texttt{List}(B)$, showing how the functorial action respects both types and functions between them.

Take $\texttt{List} : \mathbf{Set} \to \mathbf{Set}$, sending a set A to the set of lists of elements of A, and a function $f : A \to B$ to the induced function $\texttt{map}(f) : \texttt{List}(A) \to \texttt{List}(B)$. This functorial perspective captures the way type constructors preserve structure across mappings. A polymorphic function

$$\Lambda\alpha.\, t : \forall\alpha.\, F(\alpha) \to G(\alpha)$$

is then interpreted as a natural transformation $\eta : F \Rightarrow G$ between two functors $F, G : \mathscr{C} \to \mathscr{C}$. The naturality condition requires that for every morphism $f : A \to B$ in $\mathscr{C}$, the following diagram commutes:

$$\begin{tikzcd}
F(A) \arrow[r, "\eta_A"] \arrow[d, "F(f)"'] & G(A) \arrow[d, "G(f)"] \\
F(B) \arrow[r, "\eta_B"'] & G(B)
\end{tikzcd}$$

This expresses the uniformity of polymorphism: the function cannot exploit the particular structure of any type A, but must behave consistently with respect to all structure-preserving maps between types. In practice, this explains why polymorphic definitions enjoy parametricity and why they give rise to *"theorems for free."*. See Wadler 1989.

- Dependent types take polymorphism one step further. While polymorphism allows terms to be parameterized by types, dependent types allow types themselves to be parameterized by terms. This extends the Curry–Howard correspondence from propositional logic, where types correspond to propositions, to predicate logic, where types can vary with values. The dependent product type

$$\prod_{x:A} B(x)$$

generalizes the ordinary function type $A \to B$: the codomain $B(x)$ may change depending on the particular argument $x : A$. For example, one can define the type of vectors of length n, written $\texttt{Vec}(n)$, where the type explicitly depends on the natural number n. A function like vector concatenation then has the precise type

$$\forall m, n.\, \texttt{Vec}(m) \to \texttt{Vec}(n) \to \texttt{Vec}(m+n).$$

Categorically, the simply typed λ-calculus is modeled by cartesian closed categories, which provide function spaces for interpreting ordinary function types $A \to B$. To model dependent types, we need a stronger setting: locally cartesian closed categories. The key step to develop an intuition is to understand slice categories:

Definition 5.2.1. Given a category $\mathscr{C}$ and an object $A \in \mathscr{C}$, the slice category $\mathscr{C}/A$ is defined as follows:

- Objects are morphisms $f : X \to A$ in $\mathscr{C}$.
- Morphisms from $f : X \to A$ to $g : Y \to A$ are arrows $h : X \to Y$ in $\mathscr{C}$ such that $g \circ h = f$.

Intuitively, objects of $\mathscr{C}/A$ represent families of objects indexed by elements of A. For example, in **Set**, an object of $\mathbf{Set}/A$ is a function $X \to A$, which can be understood as an A-indexed family of sets. This matches the intuition of a dependent type: a type that may vary depending on the term of type A we are given.

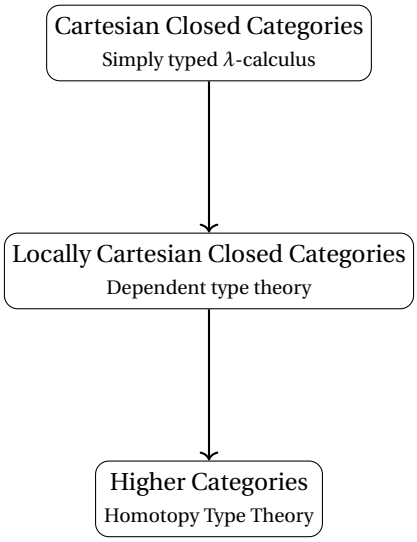
Definition 5.2.2. A category $\mathscr{C}$ is called locally cartesian closed category if every slice category $\mathscr{C}/A$ is cartesian closed for all objects $A \in \mathscr{C}$.

This extra structure means that not only do we have function spaces globally as in CCCs, but also relative function spaces within each context A. In type theory, this corresponds exactly to dependent function Π-types.

- Homotopy Type Theory builds on dependent type theory by adding ideas from homotopy and higher-dimensional mathematics. In this setting, identity types are understood as paths between points, and equalities between equalities correspond to homotopies, paths between paths. This way, types behave like spaces, with their structure reflecting not just elements but also the ways in which elements can be connected. The univalence principle says that equivalent types can be treated as identical, while higher inductive types let us define spaces directly inside type theory.

  Categorically, Homotopy Type Theory is related to the mathematics of higher categories and higher toposes, which provide abstract settings where spaces and homotopies can be studied systematically. In this sense, HoTT can be seen as extending the Curry–Howard–Lambek correspondence: not only are programs proofs and types propositions, but now types also behave like spaces, and reasoning about them corresponds to reasoning in homotopy theory.

As covered in this work, Cartesian Closed Categories provide the semantics of the simply typed λ-calculus. What follows are: Locally Cartesian Closed Categories, that extend CCCs to interpret dependent type theory. And at the next level, Higher Categories, that support the semantics of Homotopy Type Theory. Essentially:



References and Resources. For further study we relied on Nederpelt and Geuvers 2014 for an introduction to polymorphic and dependent typing, on Nordström, Petersson, and Smith 1990 for Martin-Löf's type theory, and on Jacobs 1999 for categorical semantics, including fibrations and higher-order dependent type theory. These references were not explored in full detail, but served to identify natural next steps beyond the scope of this work. Languages such as Lean 4 and other theorem provers rely on dependent type theory as their theoretical foundation. An important modern development is Homotopy Type Theory with The Univalent Foundations Program 2013, which extends dependent type theory with higher-dimensional and homotopical structure, and can be studied within proof assistants like Coq and Agda.

End.

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